Pre-commercial Procurement (PCP) FABULOS

Future Autonomous Bus Urban Level Operation System

TENDER DOCUMENT 1:

REQUEST FOR TENDERS

Deadline for receipt of the offers:

Wednesday 31 October 2018 at 17:00 PM CET

This Request for Tenders, designated as Tender Document 1, should be read in conjunction with other documents related to this Pre-Commercial Procurement (PCP), listed hereunder:

- Tender Document 2: The Functional Specification
- Tender Document 3: The Framework Agreement
- Tender Document 4: The Specific Contract for Phase 1
- Forms A through G

To submit an eligible Tender, the Tenderer shall sign and submit the Forms to the Request for Tender. The use of these Forms is mandatory.

All documents can be downloaded from and uploaded to the FABULOS website fabulos.eu

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Preface

This Request for Tenders invites all interested parties, e.g. companies and researchers, to present their vision for a smart system capable of operating a fleet of self-driving minibuses in urban environments. This means, providing a financial grant to companies to develop an all-inclusive turnkey solution for the operations of an autonomous bus line, i.e. software, hardware, fleet and services. A solution, too, that can be tested and validated on a large-scale inside cities and has the potential for large scale sustainable deployment after the project phase.

FABULOS is a research & development (R&D) project, which takes form as a Pre-Commercial-Procurement (PCP). The PCP approach and how it differs from traditional procurement, will be explained in Section 1. This Section also provides an overview of the timeline, budget, and contracting approach. In addition, a general introduction to the procurers' team (also referred to as Procuring Partners) is provided. You can find more background information about the Procuring Partners and the cities in Appendix 2.

Section 2 introduces the overall challenge this PCP must address and the motivation behind it. It explains the different phases of the PCP and the expected outcome of each phase. Finally, Intellectual Property Right (IPR) considerations are addressed. You can find more information about the technical aspects of the challenge in the 'Tender Document 2: Functional Specification'. Section 3 explains the preconditions for submitting your Tender, and an overview of the criteria to be used in the evaluation of the Tenders. The process for the evaluation of the Tenders and how to communicate with the Procuring Partners, namely putting forward your questions, is detailed in Section 4. Section 5 explains the conditions of the contracts between the winning Suppliers and the Procuring Partners, including the monitoring process, results evaluation, and payment conditions. This is followed up by a payment schedule outlined in Section 6.

This Request for Tenders should be read in conjunction with other documents related to this Pre-Commercial Procurement (PCP), listed hereunder:

Tender Documents	Forms with Tender Documents
TD1 - Request for Tender, incl. Appendices 1 to 8	**Form A - General Tender Submission Form**
TD2 - Functional Specification	**Form B - Exclusion Criteria (declaration)**
TD3 - Framework Agreement (template)	**Form C - Selection Criteria**
TD4 - Specific Contract Phase 1 (template)	**Form D - Compliance Criteria (declaration)**
	Form E - Technical Offer
	Form F - Financial Offer and Cost Breakdown
	Form G - Financial Offer Phase 1

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Table 1. Overview of the Tender Documents and Annexes

TABLE OF CONTENTS

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

1. Glossary/Definitions

Words beginning with a capital letter have the meaning defined either in this 'Request for Tender' (TD1) or in the Framework Agreement (TD3).

Terms/Acronyms	Definition
Background	Any intellectual property rights, data, software, know-how or information, whatever its form or nature (tangible or intangible), including any attached rights such as intellectual property rights ('background IPRs') - that is held by any Buyers Group member or the Supplier prior to the award of the Framework Agreement, which is needed to perform the R&D Services or exploit the Results of the PCP.
Call-off	The procedure organised by the Lead Procurer to select the successful Supplier(s) who will participate in the next Phase of the Project under the Framework Agreement.
End of Phase Report	The written report to be submitted by the Supplier to the Lead Procurer after completion of a Phase. The End of Phase Report includes all specific deliverables, Results and Sideground generated under the Phase concerned.
Fair and Reasonable Conditions	Appropriate conditions, including financial terms or royalty-free conditions, taking into account the specific circumstances of the request for access, including in particular the actual or potential value of the Results, Sideground or Background to which access is requested and/or the scope, duration or other characteristics of the exploitation envisaged.
Foreground Intellectual Property	Any intellectual property created by either party as a result of their involvement in the FABULOS framework Agreement
Framework Agreement	The contract between the Lead Procurer and the Supplier concerning the delivery of the R&D services under this PCP, covering Phases 1 through 3.
Generated in the PCP	In activities described in the PCP framework agreement or specific contracts

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Intellectual Property	Patents, inventions (patentable or capable of registration or otherwise), trademarks, service marks, copyrights, topography rights, design rights and database rights (either registered or registerable or otherwise, and including applications for registration, renewal or extension), trade secrets and rights of confidence, trade or business names and domain names and all rights or forms of protection of a similar nature which have an equivalent effect and which may now or in the future exist anywhere in the world.
Lead Procurer	The entity within the Buyers Group, appointed to coordinate and lead the joint PCP and to award and sign the Framework Agreements and Specific Contracts for all Phases of the PCP, on behalf of the Procuring Partners. The Lead Procurer is Forum Virium Helsinki, situated in Helsinki, Finland.
Not generated in the PCP	Not generated in activities described in the PCP framework agreement or specific contracts
Offer	The proposal of the Supplier for the following Phase
Preferred Partner	An entity that is not a member of the Buyers Group, but which has a special interest in closely following the PCP and therefore has access to FABULOS project-related information, as determined by the Procuring Partners.
Procuring Partners	Buyers Group. The entities procuring the R&D services under the FABULOS project. The Procuring Partners are Forum Virium Helsinki, the municipalities of Helmond, Gjesdal and Lamia, the Ministry of Communications of Estonia and public transport operator STCP from Porto. The Lead Procurer is Forum Virium Helsinki, situated in Helsinki, Finland.
Request for Tenders	The FABULOS invitation to tender on the basis of which the Tenders for the award of the Framework Agreement and the Specific Contract for Phase 1 are submitted, and the subsequently issued invitations to tender for the Phase 2 and Phase 3.
Result	Any tangible or intangible output such as data, software, knowledge or information generated under the Framework Agreement, whatever its form or nature, whether or not it can be protected, including any intellectual property rights or other rights therein. The Results expected to be generated under the Framework Agreement are identified in the relevant Specific Contract(s).
Sideground	Any tangible or intangible output, such as data, software, knowhow or information, whatever its form or nature, including any intellectual property rights or other rights therein generated during the timespan of the Framework Agreement but which does not constitute part of the Results expected to be delivered thereunder and is needed to perform the R&D services or to exploit the Results.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Specific Contract	The Contract for each Phase of the R&D services under the Framework Agreement to be concluded between the Lead Procurer and the Supplier in addition to the Framework Agreement.
Supplier	A Tenderer that is awarded a contract to execute the R&D services.
Subcontractor	A subcontractor is a third party contributing to the provision of the services referred to in the procurement contract.
Tender	The formal and commercial bid/offer submitted by the Tenderer on the basis of the Tender Documents.
Tender Documents	The following documents on the basis of which a Tenderer submits a Tender: Request for Tender, Framework Agreement, Specific Contract and Functional Specification; and its Annexes.

Table 1. Glossary / Definitions

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Abbreviations

Abbreviation	Explanation
FABULOS	Future Autonomous Bus Urban Level Operation System. Name of this European project
IPR	Intellectual Property Rights
PCP	Pre-Commercial Procurement
PPI	Public Procurement of Innovative solutions
PSC	Procurers Steering Committee
PT	Public Transport
R&D	Research and Development
SME	Small and medium-sized enterprises
WTO GPA	World Trade Organisation Government Procurement Agreement

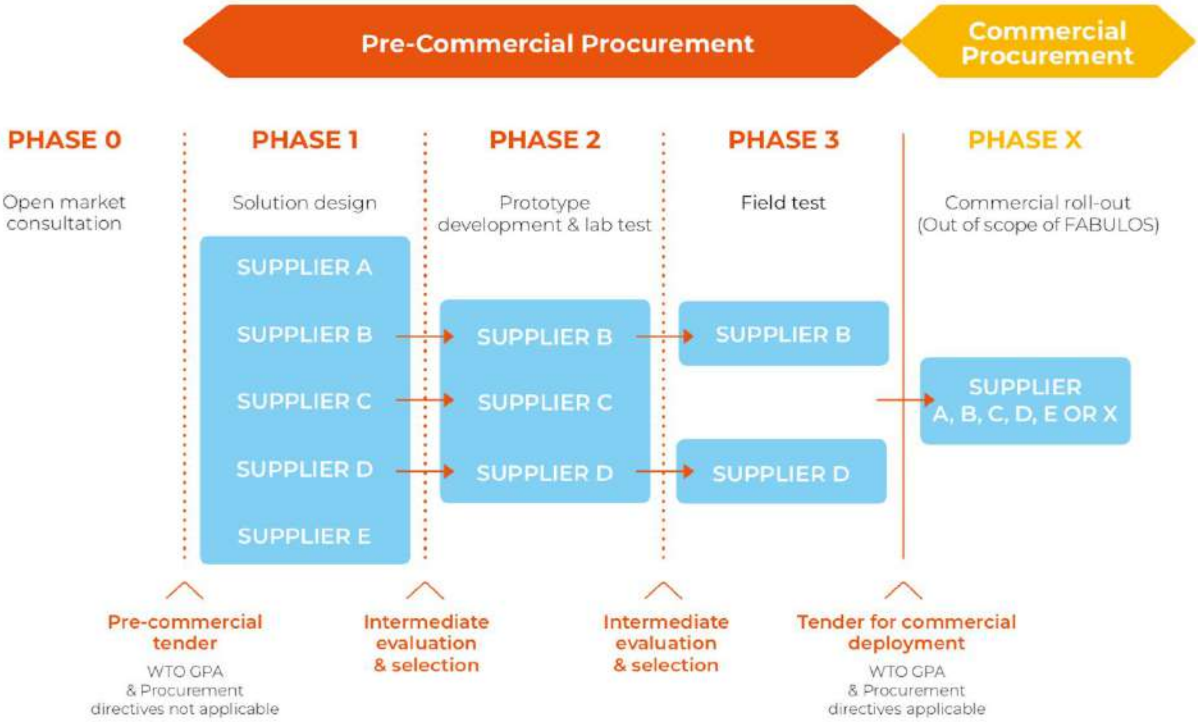
Table 2. Abbreviations

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

1.General context & background

This procurement is a **pre-commercial procurement (PCP)**.

PCP means that public procurers challenge innovative players on the market, via an open, transparent and competitive process, to develop new solutions for a technologically demanding mid- to long-term challenge that is in the public interest and requires new R&D services.



PCP is characterised by the following **features**:

Competitive development in phases to identify the solutions offering the best value for money

PCP targets situations that require radical innovation or R&D and for which there are typically no solutions on or close to the market yet. Different competing providers may have different ideas for solutions to the problem. As R&D is yet to take place, there is not yet any proof as to which of these potential alternative solutions would best meet customers' needs.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

PCP therefore awards R&D contracts to a number of competing suppliers at the same time, in order to compare different approaches to solving the problem. It thus offers innovators an opportunity to show how well their solution compares with others. It also allows a first customer test reference to be obtained from countries of the procurers that will test the solutions.

The R&D is split into **3 phases** (solution design, prototype development and lab test, field testing of a limited set of 'first' products or services). Evaluations after each phase progressively identify the solutions that offer the best value for money and meet the customers' needs. This phased approach allows successful suppliers to improve their offers for the next phase based on lessons learnt and feedback from procurers in the previous phase. Using a phased approach with gradually growing contract sizes per phase also makes it easier for smaller companies to participate in the PCP and enables SMEs to grow their business step-by-step with each phase.

Depending on the outcome of the PCP, procurers may or may not decide to follow-up the PCP with a public procurement to deploy the innovative solutions (PPI).

Public procurement of R&D services

PCP addresses mid- to long-term public procurement needs for which either no commercially stable solutions yet exist on the market, or existing solutions exhibit structural shortcomings that it requires further R&D to resolve. PCP is a way for procurers to trigger the market to develop new solutions that address these shortcomings. PCP focuses on specific identified needs and provides customer feedback to businesses from the early stages of R&D. This improves the likelihood of commercial exploitation of the newly developed solutions.

PCP is explained in the PCP communication COM/2007/799 and the associated staff working document SEC/2007/1668. The R&D services can cover research and development activities ranging from solution exploration and design, to prototyping, right through to the original development of a limited set of 'first' products or services in the form of a test series. Original development of a first product or service may include limited production or supply in order to incorporate the results of field-testing and demonstrate that the product or service is suitable for production or supply in quantity to acceptable quality standards. R&D does not include quantity production or supply to establish the commercial viability or to recover R&D costs.[1] It also excludes commercial development activities such as incremental adaptations or routine or periodic changes to existing products, services, production lines, processes or other operations in progress, even if such changes may constitute improvements.

Open, transparent, non-discriminatory approach — No large-scale deployments

PCP is open to all operators on equal terms, regardless of the size, geographical location or governance structure. There is, however, a place of performance requirement that they must perform a predefined minimum percentage of the contracted R&D services in EU Member States or Horizon 2020 associated countries. In the case of FABULOS this is 50%.

Any subsequent public procurement of innovative solutions (PPI), for the supply of commercial volumes of the solutions, will be carried out under a separate procurement procedure. Providers that did not take part in this PCP (or were not chosen to go through as far as the last phase) will thus still be able to compete on an equal basis in any subsequent procurement looking for suppliers to provide a solution on a commercial scale.

[1] See also Article XV(1)(e) WTO GPA 1994 and the Article XIII(1)(f) of the revised WTO GPA 2014.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Sharing of IPR-related risks and benefits under market conditions

PCP procures R&D services at market price, thus providing suppliers with a transparent, competitive and reliable source of financing for the early stages of their research and development. Giving each supplier the ownership of the IPRs attached to the results it generates during the PCP means that they can widely exploit the newly developed solutions commercially. In return, the tendered price must contain a financial compensation for keeping the IPR ownership compared to the case where the IPRs would be transferred to the procurers (the tendered price must be the 'non-exclusive development price'). Moreover, the procurers must receive rights to use the R&D results for internal use and licensing rights subject to certain conditions.

For more information, *see PCP on the Europa website*.

Exemption from EU public procurement directives, the WTO Government Procurement Agreement (GPA) and EU state aid rules

PCP procurements are exempted from the **EU public procurement directives** because the procurers do not retain all the benefits of the R&D (the IPR ownership stays with the suppliers).[2]

They are also exempted from the **WTO Government Procurement Agreement (GPA)** because this Agreement does not cover R&D services[3] (the PCP being limited to such services — and any subsequent PPI procurements relating to commercial-scale supply of such solutions not being part of the PCP procurement).

PCP procurements do not constitute state aid under the **EU state aid rules**[4] if they are implemented as defined in the PCP communication[5], namely by following an open, transparent, competitive procedure with risk- and benefit-sharing at market price. (The division of all rights and obligations *(including IPRs)* and the selection and award criteria for all phases must be published at the outset; the PCP must be limited to R&D services and clearly separated from any potential follow-up PPI procurements; PCP suppliers may not be given any preferential treatment in a subsequent procurement for provision of the final products or services on a commercial scale.)

Open market consultation

The start of this PCP procurement was preceded by an open market consultation (see summary and Q&A on the website fabulos.eu)

EU funding

This PCP procurement is part of a project that is funded by the European Union's Horizon 2020 Research and Innovation Programme, under grant agreement No 780371 — FABULOS (see fabulos.eu).

The contracts will therefore be subject to additional rules that come from the EU grant(s).

---

2 See Article 16(f) of Directive 2004/18/EC (Article 14 of Directive 2014/24/EU), Article 24(e) of Directive 2004/17/EC (Article 32 of Directive 2014/25/EU) and Article 13(f)(j) of Directive 2009/81/EC.
3 See the EU's Annex IV of Appendix I to the WTO GPA.
4 See Point 33 of the Commission Communication on a framework for state aid for research and development and innovation (C(2014) 3282).
5 Commission Communication: Pre-Commercial Procurement: driving innovation to ensure sustainable, high quality public services (COM(2007) 799) and PCP staff working document (SEC(2007)1668).

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

For more information, *see 'innovation procurement' and 'links to regional policy' in the [Participant Portal Online Manual](#)*.

The EU is not participating as a contracting authority in this procurement.

2. Tender profile

2.1 DESCRIPTION OF SERVICES TO BE PROCURED

PCP challenge

This procurement is for **R&D services** to develop **solutions** to tackle the following **challenge**:

The common challenge of the FABULOS PCP is the design, development and testing of smart systems that are capable of operating a fleet of self-driving minibuses in urban environments. This means, providing a financial grant to companies to develop an all-inclusive turnkey service for the operations of an autonomous bus line, i.e. software, hardware, fleet and services. A solution, too, that can be tested and validated on a large-scale inside cities, with the potential for large scale sustainable deployment after the project phase.

This is a common challenge shared by all Procuring Partners.

Many European cities are relatively well-equipped to deal with automated vehicles. Self-driving minibuses have already been tested in technical demonstrations in various European countries, among which four of the six FABULOS procuring partners. European cities can become exemplary cases to the world, but there currently is a lack of valid test cases: a proof-of-concept for the management of automated fleets as part of the public transportation provision is not yet available. This kind of intelligent transportation system and integrated transportation approach is key to facilitating the sustainable development of public transportation and for cities to be able to significantly reduce emissions, have a well-balanced mobility system based on the concept of co-modality, and foster the general longer-term goal of sustainable and liveable cities. The outcome of FABULOS will provide a commercially viable proof-of-concept, after the pre-commercial procurement process ends in December 2020.

Although other use cases are certainly imaginable, it is likely that the proposed solution mainly caters to "last mile" transport. The ends and gaps of the mobility chain, usually spanning over lengths of 0,5 - 2 km, is traditionally the most expensive part to cover with public transportation. This part of the mobility chain is often called "last mile" and with automated on-demand systems, this last mile could be covered with public transportation. Last (and first) mile play a key role when it comes to the reachability of public transport trunk network, since that is the psychologically most challenging part of the mobility chain for the passenger - without proper last/first mile solutions, the demand for passenger cars will remain high.

With a total budget of 5,442,500 Euro (incl. VAT) for supplying companies, the FABULOS Procuring Partners will purchase R&D for a turn-key solution that answers to the common

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

challenge: the design, development and testing of smart systems that are capable of operating a fleet of self-driving minibuses in urban environments. Suppliers that pass through all three Phases can receive a budget of over one million Euros.

The main quality/efficiency improvements sought for:

- The procurement target is the operation of the autonomous bus line. The procurement of a solution (service-type procurement) instead of software (product-type procurement) is needed to create an analogous model with how the public transport authorities are currently procuring the (non-autonomous) bus lines. It is expected to be similar model to the future procurements of commercial autonomous bus lines operations. This means for example that in future the cities can procure traditional bus line operations for some districts and zones as before, but after the project, in addition also will be able to procure bus transport service in zones or lines that would be managed with autonomous minibuses instead. This is an unmet need of the procuring cities.

- For cities, the autonomous minibuses are expected to lead to substantial cost savings and service level improvements, and new ability to tackle some otherwise too expensive urban transport needs. The driver is nearly always the most expensive component of the road-based public transport services.

The FABULOS approach will directly lead to:

- Creation of at least 3 separate companies or consortia with a proof-of-concept;
- Demonstrations of economic, technical, legal and societal aspects of operating of the autonomous bus service, all including the technical system for the fleet management, all validated in real life urban settings;
- Demonstration of a procurement approach that can be then easily “upgraded” for commercial procurement of the autonomous bus line operations, bridging the gap between field testing and real-life up-take of the services.

The potential for further uptake is large. There are several studies, for example EC and UITP studies (2016, 2015) that estimate the market sizes for coach, bus and urban transport with various means. The global market size estimates for autonomous buses within the urban transport vary from several billions to hundreds of billions of euros, with some US-based manufacturer estimates going far beyond these figures. More information on the expected size and type of the market in the six Procuring Partner cities can be found in Tender Document 2: Functional Specifications, Annex 2.

State of the Art

The number of autonomous shuttle pilots have increased rapidly over the past years. These pilots have started to draw interest in various cities, universities and private companies. However, the objectives of the pilots vary. In Europe, the autonomous shuttle pilots aim mostly to integrate themselves into the public transportation system. Switzerland is the most advanced country at the moment, as it has implemented autonomous shuttles to their public transport systems with regular timetables. However, in many aspects it cannot yet compete with regular public transport related to for examples driving speeds, mixed traffic situation or extreme weather conditions. Also, in most countries, including Switzerland, and employee has to be presented in the vehicle due to legislation. Removing this “steward” is important in order to obtain a commercially viable service.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Outside of Europe the pilots are still mostly about testing the technology and circumstances, or to investigate people's opinions on autonomous vehicles. How people accept autonomous shuttle buses, is one area that has not yet been researched well. Legislation is also something that affects the success of pilot projects and, therefore, some of these pilots are carried out in private areas, where the conditions are rarely comparable to real scenarios.

As part of the European SOHJOA Baltic project[6], an overview was made on past, existing and future pilots with autonomous buses, performed in open public roads with passengers. It shows that most autonomous shuttles operate with a speed of 12 km/h, with some reaching 20 km/h. In pilots with higher speeds than this, the buses are on dedicated or separate lanes: a combination of mixed traffic and speeds of 30, 40 or even 50km/hr - common in urban areas - has not yet been piloted. Mainly for safety reasons. It should be noted that the overall documentation of the autonomous shuttle pilots is insufficient to provide a complete framework.[7]

Envisaged progress beyond the State of the Art:

- A "one-stop-shop" or "off-the-shelf" solution for Cities, Public Transport Authorities or other entities procuring public transport services for the future;
- A full "last mile solution" that can compete with existing public transport (PT) but does not have such (low) velocity that it competes with walking and cycling;
- A solution that will make last-mile public transport more flexible, clean and efficient. Last (and first) mile transport plays a key role when it comes to the reachability of public transport trunk network – without proper last/first mile solutions, demand for passenger cars will remain high;
- A solution that will stimulate co-modality, including the real-time information of the location of trunk lines (rail, other buses, trams, metro) in order to provide seamless modality changes;
- A solution that can easily be fully integrated into any public transportation ecosystem. For example, integration of the fleet management to the cities' digital journey planners to both enable providing both real-time information guidance for the users, as well as to give end-users opportunity to call for on-demand autonomous bus leg as part of the mobility journey. Integrated into the cities' existing user interfaces;
- A solution that is easily scalable. This is critical to the success of both smart cities and the Suppliers of the service;
- A public transport solution that can drive in regular speeds, without an operator or steward on board, in mixed traffic in cities, while dealing with unexpected obstacles, in all weather conditions.

[6] Project website: www.sohjoabaltic.eu/en/

[7] Ainsalu, J.; Arffman, V.; Bellone, M.; Ellner, M.; Haapamäki, T.; Haavisto, N.; Josefson, E.; Ismailogullari, A.; Pilli-Sihvola, E.; Madland, O.; Madzulaitis, R.; Muur, J.; Mäkinen, S.; Nousiainen, V.; Rutanen, E.; Sahala, S.; Schønfeldt, B.; Smolnicki, P.; Soe, R.; Sääski, J.; Szymańska, M.; Vaskinn, I.; Åman, M. State of the Art of Automated Buses. *Preprints* **2018**, 2018070218 (doi: 10.20944/preprints201807.0218.v2).

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

2.2 DESCRIPTION OF THE PHASES

2.2.1 Phase 1: Solution Design

In this Phase, the Procuring Partners wish to obtain a view on how the Suppliers conceptualise the solution. Suppliers are to perform research to 1) elaborate the solution design and determine the approach to be taken to develop the new solutions and 2) demonstrate the technical, financial and commercial feasibility of the proposed concepts and approach to meet the procurement need.

Suppliers shall describe the prototype in more detail, provide a Concept Design report which includes descriptions on how the solution meets the criteria mentioned in the functional and non-functional requirements and a description of how the solution is innovative. An End of Phase Report shows among others the business model and the go to market / commercialisation plan as well as the overall project plan (for phases 2 and 3) and specific plan for phase two.

Suppliers shall also give a detailed presentation to the technical representatives of the Procuring Partners on the design and the work that was done.

The Phase lasts 3 months. The Procuring Partners intend to sign contracts with at least 10 Suppliers.

2.2.2 Phase 2 : Prototyping and lab testing

The second phase of the PCP Process is Prototyping and lab testing, consisting of the development of the concept into a working prototype. Functionality, interoperability and security tests will be performed in a lab environment, simulating urban conditions. This Phase can be executed with one vehicle only, and it does not require an entire fleet.

In Phase 2, the prototype development, lab testing and closed street testing will be conducted in the facilities of the selected Suppliers. There will be an intermediate validation in the lab testing phase and an interim payment upon approval. To monitor the progress and facilitate the dialogue, the phase will also have a series of iterations, where the supplier presents the current status of the prototype to the procuring partners, and receives feedback on the current status.

The functional requirements and key features to be tested include: Safety and security, Call-to-order, Fleet management and Control room functions. The Traffic Safety Validation includes validating that the vehicle is able to move safely in simulated urban conditions, on its own power and using its on-board sensors. This phase will provide technical feedback to the participating Suppliers. Outcome of the phase will be an Evaluation report together with a detailed plan for the pilot, functioning prototype and test demonstration of the prototype in lab conditions.

After receiving approval of the intermediate evaluation in Phase 2, Suppliers already need to apply for exemptions and permissions in the six Procuring partner cities, as these processes can take much time. By the end of Phase 2, permissions and exemptions need to be in place for Phase 3 field tests.

At the end of the Phase, the Suppliers shall also provide an End of Phase Report and give a detailed presentation to the technical representatives from the Procuring Partners on the pilot solution. The End of Phase Report shall be a written documentation of the prototype including

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

development details, operational procedures, specifications report, test demonstration of the prototype in lab conditions, updated cost/benefit evaluation and description of the result, conclusions and the development plan for the continuation of the development activities in Phase 3. Also, a report explaining the output and feedback from the monitoring briefings by the Procuring Partners.

More detailed information will be provided at the Phase 2 Call-off.

The Phase lasts 6,5 months. The Procuring Partners intend to sign contracts with at least 6 Suppliers.

2.2.3 Phase 3: Field Testing

In this final Phase, the prototypes are developed further into solutions that are validated in real-life settings. This translates into the prototypes' integration into field tests that are set up in urban environments in some or all of the Procuring Partner cities. The objective is to provide the Procuring Partners with a proof-of-concept and sufficient insight into the capabilities of the supplier to provide a solution which is in line with the challenge and which can work in a real-life environment.

Therefore, in Phase 3, field testing of the prototypes takes place on predetermined pilot routes in all or some of the Procuring Partner cities. The pilot will have an interim evaluation and payment.

The output is the successful completion of at least 2 field tests per Supplier. These field tests can be done in parallel or consecutively. Stable field tests that work on a technical level and can be tested in real life settings with end users, described in an End of Phase Report and demonstrated to the Procuring Partners as input towards a potential future procurement procedure after FABULOS has run its course.

The specific setup and circumstances of each field test will be further detailed, together with Suppliers, in the Phase 3 call-off stage. The currently foreseen pilot routes are described in Tender document 2: Functional Specifications (Annex 1).

In addition, for more information on the Procuring Partner cities, please see Appendix 2 of the underlying document. For each of the field testing cities you can find a description of the city, its public transport system and mobility policy and relevant data. Open API's, information on wireless network capabilities, on the vendors and technologies currently used for traffic lights and on the availability of route-wide remote camera surveillance is found on Annex 2 of the Tender Document 2: Functional Specifications, together with information about local regulations and exemption process in Procuring Partner cities for automated vehicles.

The Phase lasts 6,5 months. The Procuring Partners intend to sign contracts with at least 3 Suppliers.

Allocation of field test sites (Phase 3)

The Procuring Partners have made three pairs of two cities for the field testing in Phase 3. These pairs try to take into account price level, geography, size and application time for permissions/exemptions. In Phase 2, Suppliers can indicate their preference. After the evaluation

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

of Phase 2 and assessment of the offers for Phase 3 is done, the Procuring Partners will make a joint decision on which consortium will do its field testing in which cities.

Field testing pairs:

Gjesdal & Porto
Helsinki & Lamia
Helmond & Tallinn

2.3 EXPECTED OUTCOMES

This section describes the Milestones and Deliverables resulting from each Phase. All Milestones and Deliverables mentioned below are the responsibility of the Suppliers.

In addition, promotional materials can be requested ad hoc in all Phases for dissemination purposes (e.g. short videos/demo).

2.3.1 Phase 1 Deliverables

Milestones and deliverables		By when?	How?	Output and results
Milestone:	M1.1) Kick-off Phase 1	Start of Phase 1	Presentation (webinar)	Presentation of the supplier's team and solution + Q/A
Deliverable:	D1.1) Project abstract for Phase 1 and list of Pre-existing IP (see 3.5)	2 weeks after Phase 1 start	Document (format required by the EU for publication)	Project abstract and list of Background IP.
Milestone:	M1.2) Interim follow-up	6 weeks after start phase 1	Discussion (webinar)	Discussion on the supplier's project evolution
Milestone:	M1.3) End of Phase review	End of Phase 1 (3 months after start of Phase)	Presentation + evaluation (Physical meeting, in Helsinki*)	Evaluation of the supplier's solution
Technical Deliverable:	D1.2) Concept presentation (description in deliverable table)	End of Phase 1	Document and Presentation	End of Phase Report
Technical Deliverable:	D1.3) Concept design report (description in deliverable table)	End of Phase 1	Document and Presentation	End of Phase Report
Non-technical deliverable:	D1.4) End of Phase report (description in deliverable table)	End of Phase 1	Document	End of Phase report
	D1.5) Phase 2 Offer	End of Phase 1	Document	Next Phase proposal

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

	Points to be addressed in end-of-phase report: see description in deliverable table and Appendix 8 for sample report.

Table 3. Outcomes of Phase 1

**) See paragraph 2.3.5. Meeting locations are tentative and will be confirmed before signing the Phase 1 contract.*

2.3.2 Phase 2 Deliverables

Milestones and deliverables		**By when?**	**How?**	**Output and results**
Milestone:	M2.1) Kick off phase 2 meeting	Start of Phase	Visit of phase 2 suppliers to the premises of one of the procuring partners (Tallinn, Estonia*)	Presentation of the supplier's team and solution + Q/A
Deliverable:	D2.1)Project abstract for Phase 2 and updated list of pre-existing IP	two weeks after Phase 2 start	Document (format required by the EU for publication)	Project abstract and list of Background IP
Milestone:	M2.2) First Prototype iteration Interim briefing	Six weeks after the start of phase two	Webinar	Presentation of the supplier's project evolution (testing in own lab)
Milestone:	M2.3) Second Prototype iteration and lab test demonstration	3,5 months after start of Phase 2	Demonstration. Evaluation team to visit suppliers' labs	Demonstration of lab test results
Technical Deliverable:	D2.2) Lab test results report	3,5 months after start of Phase 2	Document	Report on the current status of the Prototype and lab test results. Based on satisfactory results M2.3 and D2.2: interim payment and green light to start applying permissions and exemptions for Phase 3
Milestone:	M2.4) Third (final) Prototype iteration Interim briefing	5 months after start of phase 2	Webinar	Presentation of the supplier's project evolution (closed street testing and preparations for

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

				testing on open streets)
Milestone:	M2.5) Verification of prototype testing on closed streets. Check completion of milestone(s) / deliverable(s)	End of Phase 2 (6,5 months after start of Phase)	Demonstration. Physical meeting and presentation Visual demonstration of the solutions' prototype in City of Helmond or in Suppliers' location of choice *)	Demonstration of prototype in closed street environment
Technical Deliverable:	D2.3) Final report on prototype technical functionalities covering entire Phase 2	End of Phase 2 6,5 months after start of Phase)	Document	Presentation and report, including Prototype lab test results and on street prototype testing results.
Non-technical Deliverable:	D2.4) End of Phase report (description in deliverable table)	End of Phase 2 6,5 months after start of Phase)	Document (upload on the online platform)	End of Phase report
	D2.5) Permissions and exemptions in place for Phase 3 field tests	End of Phase 2 6,5 months after start of Phase)	Document	Permissions and exemptions in place for Phase 3 field tests
	D2.6) Phase 3 offer	End of Phase 2	Document	Next phase proposal
Points to be addressed in end-of-phase report:	Description of the state-of-the-art versus innovation gap Measures taken to protect Results (IPR) List of names and location of personnel that carried out the R&D activities Abstract of the main Results achieved and conclusions from phase 2 (EU-format) Commercialisation plan Detailed time schedule and implementation plan field testing in phase 3 Re-assessment of the R&D efforts for the field testing See Appendix 8 for sample End of Phase report.			

Table 4. Outcomes of Phase 2

**) See paragraph 2.3.5. Meeting locations are tentative and will be confirmed before signing the Phase 2 contract.*

2.3.3 Phase 3 Deliverables

Milestones and deliverables		**By when?**	**How?**	**Output and results**
Milestone:	M3.1) Intake meeting and Pilot preparation	Start of Phase	Visit of phase 3 suppliers to the premises of one of	Completed field test validation exercise, agreement on use cases,

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

			the procuring partners (Gjesdal, Norway*)	milestones, data policies and practical
Technical Deliverable:	D3.1) Field test Validation Exercise	Start of Phase	Physical meeting	Completed Field test Validation and goal definition
Technical Deliverable:	D3.2) Detailed Field Test work plan (feedback received during selection of phase 3 Suppliers is taken into account in the plan)	Start of Phase	Document	Concrete agreement on routes, timeplan, milestones, data policies and other practicalities
Deliverable:	D3.3) Project abstract for Phase 3 and list of Pre-existing IP (see paragraph 3.5)	2 weeks after Phase 3 start	Document (format required by the EU for publication)	Project abstract and list of Background IP
Milestone:	M3.2) First Field test monitoring (for each Field Test location)	1,5 months after start of Phase 3	Webinar	Evaluation of the first part of Field Testing and formulation of aspects that require iteration
Milestone:	M3.3) Second Field test monitoring: interim (for each Field Test location)	3 months after start of Phase 3	Physical meeting with demonstration. Evaluation team to visit suppliers' field test locations	Interim field Test evaluation: demonstration
Technical Deliverable:	D3.4) Interim Field test iteration exercise (for each Field Test location)	3 months after start of Phase 3	Report	Description of the field test progress in each city, with iteration plan for next 3 months Based on satisfactory results M3.3 and D3.4: interim payment.
Milestone:	M3.4) Third Field Test monitoring (for each Field Test location)	5 months after start of phase 3	Webinar	Presentation of the Supplier's Field Test evolution
Milestone:	M3.5) final field test monitoring (for each Field Test location)	End of Phase (6,5 months after start of Phase)	Physical meeting with demonstration Evaluation Committee and EC reviewers to visit suppliers' field test locations	Evaluation of the whole Field test phase in a presentation and report
Technical Deliverable:	D3.5) Final Field Test - Prototype iteration exercise (for each Field Test location)	End of Phase (6,5 months after start of Phase)	Report	Evaluation of the whole Field test phase in a presentation and report

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Milestone:	M3.6) Demo of final solution. End of Phase 3 review (for each Field Test location)	End of Phase (6,5 months after start of Phase)	Potential additional physical meeting. Demonstration to the EC (usually Brussels, Belgium)	Successful demonstration of the work done in the 3 Phases to EU representatives
Technical Deliverable:	D3.6) end of phase report	End of Phase (6,5 months after start of Phase)	Document	End of Phase Report
Points to be addressed in end of Phase report:	Description of the state-of-the-art versus innovation Measures taken to protect Results (IPR) List of names and location of personnel that carried out the R&D activities Abstract of the main results achieved and conclusions from the PCP in the format required by the EU for publication) Commercialisation plan See also Appendix 8 for sample End of Phase report.			

Table 5. Outcomes of Phase 2

Milestones and Deliverables for the Phases 2 and 3 are indicative at this stage and could be subject to change. Any changes will be included in the respective Phase's call-off stage.

2.3.4 Detail Phase 1 Deliverables

D1.2	**Concept presentation**	• Showcase the prototype concept and what was realised in this phase • Showcase the prototype architecture and technical design components • Show how the prototype realises the different functional criteria • Show how the prototype realises the different quality criteria • Show the business model and the go to market / commercialisation plan • Show overall project plan (for phases 2 and 3) and specific plan for phase 2
D1.3	**Concept design report**	Document that describes the prototype in more detail. To be presented in accordance with the following: • Max 2 pages per functional and nonfunctional requirement that shows how the solution will meet the must-have criteria and nice-to-have criteria. • Max one page per requirement that shows how the solution is innovative • Max one page that shows how the solution is innovative in other domains

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

D1.4	**End of Phase Report**	• Description of the state-of-the-art versus innovation gap • Measures taken to protect Results (IPR) • List of names and location of personnel that carried out the R&D activities • Abstract of the main Results achieved and conclusions from phase 1 (EU-format) • Commercialisation plan • Detailed time schedule and implementation plan • Detailed report covering all the results in the concept phase • Re-assessment of the R&D efforts for the prototype and field testing

D1.5	**Phase 2 Offer**	• Project Management plan & Project Team • Detailed Project Plan for Phases 2 and 3 • Adapted Risk Plan • Adapted Quality Assurance Plan • Impact on Challenge • Technical Quality of the solution • Quality Assurance and Testing • Commercial Feasibility, business model and marketing plan • Field testing • Intellectual Property

Table 6. Detail Phase 1 Deliverables

2.3.5 Overview of travels

This table below gives an overview of the envisaged travels that the Suppliers need to plan for in their financial offer. The timing and locations for the physical meetings in all Phases, particularly in the Phases 2 and 3, are indicative at this stage and could be subject to change. The FABULOS Procuring Partners reserve the right to adjust the duration of the iteration periods, meetings frequency and locations if necessary. This will be communicated in a timely manner to all Suppliers.

Phase	What	When	Where
Phase 1	M1.3) End of Phase review	End of Phase 1 (3,5 months after start of Phase)	Helsinki, Finland
Phase 2	M2.1) Kick off phase 2 meeting	Start of Phase	Tallinn, Estonia

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Phase 2	M2.5) Verification of prototype testing on closed streets. Presentation of complete milestones / deliverables. Demonstration of solution prototype.	End of Phase 2 (6,5 months after start of Phase 2)	Helmond, Netherlands. Alternatively, at premises Supplier. In case of the latter, Supplier needs to arrange and pay travels of Evaluation Committee.
Phase 3	M3.1) Intake meeting and Pilot preparation. Presentation of field test validation exercise, and Field test work plan, including routes, timing, milestones, data policies and practical arrangements	Start of Phase	Gjesdal, Norway
Phase 3	M3.5) Final field test monitoring (for each Field Test location) Physical meeting with demonstration.	End of Phase (6,5 months after start of Phase 3)	Evaluation Committee and EC reviewers to visit suppliers' field test locations. Suppliers need to arrange and pay travels of 2 EC representatives to their field test sites.
Phase 3 (optional)	Demonstration during TRA 2020 in Helsinki. Voluntary for Suppliers that are not field testing in Helsinki.	26-30.4.2020	Helsinki fair centre. The (planned) field testing route passes the fair centre.

Table 7. Overview of Travels

2.4 EVALUATION AND PROCESS OVERVIEW

2.4.1 End of Phase 1

The FABULOS consortium will provide a document to all selected Suppliers before the start of Phase 1 that elaborates on the Phase 1 Specific Deliverables and Evaluation. It provides guidelines to the Suppliers in order to prepare for a successful delivery of Phase 1 Results and the consequent Evaluation process. The Guideline document on Phase 1 Specific Deliverables and Evaluation will a.o. include a timing overview, a description of the evaluation process (interim monitoring and end of phase review), the collaboration tools the FABULOS and the suppliers will use and all templates for reports.

The FABULOS Phase 1 evaluation process is intended to evaluate the Solution Designs as proposed by the selected Suppliers for Phase 1. The evaluation process is 2-phased:
1) Interim Project Monitoring via webinar +/- 6 weeks after the start of Phase 1. This monitoring is intended to gain an interim insight in the Suppliers' progress and provide the opportunity for interaction between the Suppliers and the Procuring Partners on the solution design. The

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

interim
monitoring will not be scored;
2) The End-of-Phase evaluation, 3,5 months after the start of Phase 1. This evaluation is intended to assess and score the developed solutions. The End-of-Phase evaluation will decide upon the Satisfactory and/or Successful completion of Phase 1.

Solutions will be evaluated in a non-discriminatory and transparent manner. In order to achieve this, the FABULOS project structure has foreseen in a Technical Evaluation Committee and an independent Advisory Evaluation Panel. The Procuring Partners will evaluate the technical and non-technical milestones and deliverables comprised in the End of Phase 1 Report. All Milestones and Deliverables will be scored according to the stipulations in TD1 - Conditions of the contracts, TD2 Functional Specification, and the Scoring Model for Outcome of the Phase(s) in Appendix A. The weights for the evaluation of the Phase 1 are the same as those of the Contract Awarding.

A consolidated End-of-Phase Evaluation Report and a final Supplier ranking will be approved by the Procurers Steering Committee and will be delivered to the European Commission.

All competing Suppliers will receive the call-off for Phase 2 and are expected to provide a Phase 2 offer based on these call-off documents. However, the **successful** completion of Phase 1, including the final report validation and Solution Design approval, is a **prerequisite** to have your Phase 2 tender evaluated.

Payments corresponding to each PCP phase will be subject to the **satisfactory** completion of the deliverables and milestones for that phase. Satisfactory completion will be assessed by the Technical Evaluation Committee . They will take the final decision on the acceptance or rejection of the milestones/deliverables/tests. Satisfactory completion in each of the phases does not mean successful completion. For more explanation on and the difference between **satisfactory** completion and **successful** completion, please see section 5.5 of this document.

2.4.2 End of Phase 2

In Phase 2 the Contractors are asked to provide a prototype for the solution that will address the FABULOS challenge and all functional and quality requirements to enable the implementation of a proof-of-concept on autonomous last mile public transport as part of the urban areas' existing transport system. On the basis of the Functional Specification (TD2), the goal is to demonstrate the technical, financial, legal and commercial feasibility of the proposed concepts and approach to meet the procurement needs. The evaluators will note progress based upon actual information and evidence from the FABULOS project, not generalities and possibilities. All Phase 2 Deliverables must be provided to FABULOS at a maturity level that exceeds the level of the previous phase and in a structured way.

Also for the end of Phase 2, the FABULOS consortium will provide Suppliers with a document that elaborates on the Phase 2 Specific Contract Milestones and Deliverables and provides guidelines to the Suppliers to prepare for a successful delivery of the Phase 2 results and the consequent Evaluation process. It will contain dates and details on the Iterations and interim monitoring, an overview and templates of Milestones and Deliverables for Phase 2, the exact

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Phase 2 timeline, available collaboration tools, information on the cities' expectations, adapted testing and reporting requirements (in case of changes compared to the Call off documents for Phase 2), a section on Dissemination and on the evaluation process.

The End-of-Phase evaluation is intended to assess and score the developed prototypes, their lab testing and closed street testing. The End-of-Phase evaluation will decide upon the Satisfactory and/or Successful completion of Phase 2. The work will be evaluated in a non-discriminatory and transparent manner. Failing or succeeding to deliver the required Deliverables and Results will be a part of the Phase 2 evaluation process. A consolidated End-of-Phase Evaluation Report and a final Supplier scoring result will be approved by the Procurers Steering Committee and will be delivered to the European Commission. All competing Phase 2 Suppliers will receive the call-off for Phase 3 and are expected to submit and offer for Phase 3 based on these call-off documents. However, the successful completion of Phase 2, including the final report validation and Solution Design approval, is a prerequisite to have your Phase 3 tender evaluated.

On the end date of Phase 2 the Suppliers shall submit their 'End of Phase Report', including all the listed deliverables and Results. The End of Phase report will be assessed to determine whether the respective Suppliers have satisfactorily or unsatisfactorily completed a Phase. The satisfactory/unsatisfactory completion will determine whether the Supplier shall be paid or not. The Evaluation Committee will also evaluate, if the Suppliers have successfully completed the particular Phase, and consequently, if they are eligible to participate in the call for tender for Phase 3. Successful completion of a Phase will be assessed by the Evaluation Committee against the End of Phase Report (including all deliverables) according to the evaluation criteria, the scoring table in TD1 - Appendix 6 and under the condition that the key milestones have been successfully completed.

2.4.3 End of Phase 3

Also for the end of Phase 3, the FABULOS consortium will provide Suppliers with a document that elaborates on the Phase 3 Specific Contract Milestones and Deliverables and provides guidelines to the Suppliers to prepare for a successful delivery of the Phase 3 results and the consequent Evaluation process. It will contain dates and details on the Iterations and interim monitoring, an overview and templates of Milestones and Deliverables for Phase 3, the exact Phase 3 timeline, available collaboration tools, information on the cities' expectations, adapted testing and reporting requirements (in case of changes compared to the Call off documents for Phase 3), a section on Dissemination and on the evaluation process.

The End-of-Phase evaluation is intended to assess and score the developed prototypes. The End-of-Phase evaluation will decide upon the Satisfactory and/or Successful completion of Phase 3. The work will be evaluated in a non-discriminatory and transparent manner. Failing or succeeding to deliver the required Deliverables and Results will be a part of the Phase 3 evaluation process. A consolidated End-of-Phase Evaluation Report and a final Supplier scoring result will be approved by the Procurers Steering Committee and will be delivered to the European Commission.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

2.5 PROCURERS AND OTHER PARTIES INVOLVED IN THE PCP

This procurement relates to a joint PCP that will be carried out by the following **lead procurer**: Forum Virium Helsinki, Finland. The lead procurer is appointed to coordinate and lead the joint PCP, and to sign and award the framework agreement and the specific contracts for all phases of the PCP, in the name and on behalf of the following **Procuring Partners**. The Procuring Partners are the group of procurers that contribute to the procurement budget.

- MAJANDUS JA KOMMUNIKATSIOONIMINISTEERIUM, Estonia, hereinafter referred to as 'MKM'
- DIMOS LAMIA, Greece, hereinafter referred to as 'Lamia'
- SOCIEDADE DE TRANSPORTES COLECTIVOS DO PORTO SA, Portugal, hereinafter referred to as 'STCP'
- GEMEENTE HELMOND, Netherlands, hereinafter referred to as 'Helmond'
- GJESDAL KOMMUNE, Norway, hereinafter referred to as 'Gjesdal'

The lead procurer is part of the Procuring Partners.

Each member of the Procuring Partners has one representative and voting right in the Procurers Steering Committee (PSC). The PSC will validate all key steps to be taken in the preparation of the request for Tenders, and in the performance of the 3 stages of the PCP process. This includes the final validation of the specific challenge in the request for tender, and the validation of the successful tenderers to be awarded a contract in each of the three phases, including any decisions regarding the non-satisfactory completion of any specific phase by a supplier.

In Appendix 2, the background and responsibilities of the Procuring Partners in relation to for example their role with regard to public transport, mobility policy and open data are described.

To ensure the further validation of the pre-commercial procurement challenge definition, and more direct future market up-take for the solution, the FABULOS project has set up a "Preferred partner" group consisting of cities and/or public procurers. The following entities are participating as preferred partners with an interest in the PCP, but without being part of the buyers group or giving in-kind contributions for carrying out the PCP:
City of Copenhagen (DK)
MOVIA - the regional transportation company for the Greater Copenhagen region (DK),
City of Ghent (BE)
City of Trikala (GR)
City of Delphi (GR)
City of Torino(IT)
City of Fundão (PT)
City of Kongsberg (NO)
City of Velje (DK)
City of Vantaa (FI)
Region of Stavanger (NO)
Porto Digital (PT)
City of Tuusula (FI)
City of Buenos Aires (AR)

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

The Preferred Partners provide information and opinions to help the procuring partners' decision-making, for example, by participating in some of the activities of the Open Market Consultation. They promote the results of the FABULOS pre-commercial procurement process to the policy community and other adopter cities. They also promote the FABULOS request for tenders to potential suppliers in their constituencies. They will keep the FABULOS partners informed of any implementation of self-driving buses in their municipalities and the results..

Preferred Partners are entities that are neither members of the buyers group, nor third parties providing in-kind contributions to the FABULOS project, but that have a special interest in closely following the FABULOS pre-commercial procurement. They are interested in implementing the solution coming out of FABULOS in their own cities. They will not have rights to results or IPRs. The Preferred partners will receive some funding to cover travel costs up to two FABULOS events, of which one is the Final conference. During the Final Conference demonstrations of the final solutions will be given. A special workshop session for Preferred Partners will be hosted, with a view to a potential common PPI.

Also, a FABULOS Preferred Partners event will be organised in the margins of the Transport Research Arena Conference in April 2020 in Helsinki. This will also be the time that the pilot in Helsinki is ongoing, so Preferred Partners will get a demonstration of the solution and a peek behind the scenes.

2.6 CONTRACTING APPROACH

The PCP will be implemented by means of a **framework agreement** with **specific contracts** for each of the 3 R&D phases (altogether 'contracts'). The law governing the contracts is the Finnish law, because the Lead Procurer is based there.

Following the tendering stage, a framework agreement and a specific contract for phase 1 will be awarded to an expected minimum of 10 suppliers.

A call-off will be organised for phase 2, with the aim of awarding an expected minimum of 6 phase 2 specific contracts. Only offers from suppliers that successfully completed phase 1 will be eligible for phase 2. The procurers will validate the phase 2 prototypes.

A second call-off will be organised for phase 3, with the aim of awarding an expected minimum of 3 phase 3 specific contracts. Only offers from suppliers that successfully completed phase 2 will be eligible for phase 3. Phase 3 field-testing is expected to take place, per Supplier, in 2 of the following cities: Helsinki (Finland), Tallinn (Estonia), Helmond (Netherlands), Gjesdal (Norway), Lamia (Greece), Porto (Portugal).

The framework agreement will set all the framework conditions for the entire duration of the PCP (covering all the phases). There will be no renegotiation. The framework agreement will remain binding for the duration of all phases for which suppliers remain in the PCP. Suppliers that are awarded a framework agreement will also be awarded a specific contract for phase 1 (evaluation of tenders for the framework agreement and phase 1 are combined). Suppliers therefore need not only to submit their detailed offer for phase 1, but also to state their goals, and to outline their plans *(including price conditions)* for phases 2 and 3 — thus giving specific details of the steps that would lead to commercial exploitation of the R&D results.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

The offers for the next phase will be requested together with the end-of phase deliverables for the previous phase. However, the successful completion of the previous phase is evaluated before evaluating the offers for the next phase, to determine which offers are eligible to proceed to the evaluation of offers for the next phase. Consequently, if a supplier's phase results is not consider successful, its offer for next phase will not be evaluated.

Short overview of timing:

Phase 1: January - March 2019 (3 months)

Phase 2: June - December 2019 (6,5 months)

Phase 3: March - September 2020 (6,5 months)

A more detailed time schedule can be found in paragraph 2.8.

2.7 TOTAL BUDGET AND BUDGET DISTRIBUTION

The total budget for the PCP is 5.442.500 € incl 24% VAT and 4.136.300 € excl. 24% VAT. Suppliers that participate in all 3 Phases can receive a total budget of up to € 1.046.280.

Expected minimum number of Suppliers	Max. budget per phase, excl. 24% VAT (indication)	Max. budget per supplier, excl. 24% VAT (indication)	Max. duration of phase
Phase 1: 10	€ 346.560	€ 34.656	3 months
Phase 2: 6	€ 1.509.738	€ 251.623	6,5 months
Phase 3: 3	€ 2.280.002	€ 760.001	6,5 months
Total	*€ 4.136.300*	*€ 1.046.280*	

Table 8. Overview of budgets per Phase

For phases 1 and 2, contracts will be financed until the remaining budget is insufficient to fund the next best tender. The exact number of contracts finally awarded will thus depend on the prices offered and the number of tenders passing the evaluation.

As leftover budget from the previous phase will be transferred to the next phase, the total budget available for phases 2 and 3 may eventually be higher than stated here (but the maximum budget per supplier for phases 2 and 3 will in principle remain the same). The lower the average price of tenders, the more Suppliers can be awarded. However, the total value of the contracts awarded can also be lower than initially expected if there are fewer tenders than expected that meet the minimum evaluation criteria.

Since all suppliers will be paid by the Lead Procurer (centralised payments) and as Forum Virium Helsinki is the Lead Procurer in the FABULOS PCP, the valid Finnish and EU VAT legislation will be applied in the project.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

2.8 TIME SCHEDULE INCLUDING TENDER CLOSE

The official tender closing time will be: Wednesday 31 October 2018 at 17:00 CET

First tender procedure (framework agreement and phase 1 contracts)

Date	Activity
01.09.2018	Publication of contract notice in TED. Tender documents for download on fabulos.eu
11.09.2018	Q&A session and presentation of the Tender (webinar)
15.10.2018	Q&A overview and final clarification session (webinar)
16.10.2018	Deadline for submitting questions about tender documents via ask@fabulos.eu
21.10.2018	Deadline for lead procurer to publish replies to questions (Q&A document)
31.10.2018	Deadline for submission of tenders for the framework agreement and phase 1
1.11.2018	Opening of tenders
30.11.2018	Tenderers notified of decision on awarding contracts
17.12.2018	Signing of Framework Agreements and Phase 1 Specific Contracts
19.12.2018	Publication of contract award notice in TED

Implementation of phase 1

Date	Activity
7.1.2019	Start of phase 1 work by suppliers
9-10.1.2019	Webinars for phase 1 suppliers: one common to learn about the operational boundary conditions governing the design of the solutions. In addition, individual telcos for each winning consortium to discuss the evaluation reports (M1.1)
23.1.2019	Project abstract for Phase 2 and updated list of pre-existing IP (D1.1)
13-14.2.2019	Interim follow-up: webinar with each consortium to discuss project evolution (M1.2)
21.2.2019	Phase 2 call-off specifications to suppliers
28.2.2019	Webinar on the Phase 2 call off specifications
22.3.2019	Deadline for submitting questions on end-of-phase 1 reporting templates and phase 2 call-off specifications
27.3.2019	Deadline for lead procurer to circulate replies to questions
5.4.2019	Deadline for phase 1 final milestone(s)/final report/deliverable(s), including phase 2 offers (D1.2, D1.3, D1.4, D1.5)
8-12.4.2019	Suppliers to present results of Phase 1 to Procuring Partners in Helsinki, Finland (preliminary location) (M1.3)

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

13.5.2019	Phase 1 suppliers notified as to whether they have completed this phase satisfactorily and successfully and of decision on awarding phase 2 contracts
13.5.2019	Suppliers whose performance is considered satisfactory and successful are to submit the final invoice for Phase 1.
10.6.2019	Payment of balance for phase 1 to suppliers that completed this phase satisfactorily made as soon as notified of winners.

Second tender procedure

Date	Activity
16.4.2019	Opening of phase 2 offers
13.5.2019	Phase 1 suppliers notified as to whether they have completed this phase satisfactorily and successfully and of decision on awarding phase 2 contracts
27.5.2019	Signing of phase 2 specific contracts

Implementation of Phase 2

Date	Activity
27.5.2019	Signing of phase 2 specific contracts
1.6.2019	Start of phase 2
3-4.6.2019	Visit of phase 2 Suppliers to the premises of one of the procurers for a Start-of-Phase Meeting (M2.1)
17.6.2019	Project abstract for Phase 2 and updated list of pre-existing IP (D2.1)
15.7.2019	First prototype iteration: webinar on Suppliers' project evolution (M2.2)
16.9.2019	Deadline for phase 2 interim milestones/deliverables
18-25.9.2019	Visits of the phase 2 supervisor/monitoring team to the supplier's premises to check completion of interim milestones)deliverables (second prototype iteration) (M2.3, D2.2)
10.10.2019	Feedback from phase 2 supervisor/monitoring team on phase 2 interim milestones/deliverables
10.10.2019	Suppliers whose interim performance is considered satisfactory and successful are to submit a mid-term invoice
31.10.2019	Phase 2 Interim payments
10-11.2019	Closed street & lab testing of the prototype developed during phase 2
6.11.2019	End-of-phase reporting templates and phase 3 call-off specifications to Suppliers
7.11.2019	Third Prototype iteration: webinar on Suppliers' project evolution (M2.4)
2.12.2019	Deadline for submitting questions on end-of-phase 1 reporting templates and phase 2 call-off specifications
6.12.2019	Deadline for lead procurer to circulate replies to questions
13.12.2019	Deadline for submission of phase 2 final milestones/final report /deliverables, including phase 3 offers (D2.3, D2.4, D2.5, D2.6)
9-18.12.2019	Demonstration of Phase 2 prototype to the Evaluation Committee and EU technical reviewers. In a location in Europe to be determined by Procuring Partners. (M2.5)
6-31.1.2020	Assessment of phase 2 final milestones/final report/deliverables and phase 3 offers
22.1.2020	Opening of phase 3 offers

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

10.2.2020	Phase 2 suppliers notified as to whether they have completed this phase satisfactorily and successfully and of decision on awarding Phase 3 specific contracts
10.2.2020	Suppliers whose performance is considered satisfactory and successful are to submit the final invoice for Phase 2.
5.3.2020	Payment of balance for phase 2 to suppliers that completed this phase satisfactorily

Third tender procedure

Date	Activity
22.1.2020	Opening of Phase 3 offers
10.2.2020	Suppliers notified of decision on awarding Phase 3 contracts
20.2.2020	Signing of phase 3 specific contracts

Implementation of Phase 3

Date	Activity
1.3.2020	Start of phase 3
4-5.3.2020	Intake meeting: Visit of phase 3 suppliers to premises of one of the procuring partners. Field test validation exercise and detailed field Test workplan (M3.1, D3.1, D3.2)
19.3.2020	Project abstract for Phase 3 and updated list of pre-existing IP (D3.3)
15.4.2020	First Field test monitoring webinar (M3.2)
26-30.4.2020	Possibility for Demonstration during TRA 2020 in Helsinki (optional for Suppliers that are not field testing in Helsinki)
5.6.2020	Deadline for phase 3 interim milestones/deliverables (D3.4)
8-17.6.2020	Interim field test monitoring. Visits of the phase 3 Evaluation Committee to the pilot locations to check completion of phase 3 interim milestones/deliverables (M3.3)
30.6.2020	Feedback from phase 3 monitoring supervisor/monitoring team on phase 3 interim milestones/deliverables
30.6.2020	Suppliers whose interim performance is considered satisfactory and successful are to submit a mid-term invoice
17.7.2020	Phase 3 Interim payments
30.7.2020	Field test monitoring webinar (M3.4)
15.09.2020	Deadline for submission of phase 3 final milestones/final report/ deliverables
9 - 18.9.2020	Final demonstration of solution developed during phase 3 (including to EU representatives). Evaluation team to visit pilots
21.9-19.10.2020	Assessment of phase 3 final milestones/final report/deliverables
5.11.2020	Phase 3 suppliers notified as to whether they have completed this phase satisfactorily and successfully
5.11.2020	Suppliers whose performance is considered satisfactory and successful are to submit the final invoice for Phase 3.
3.12.2020	Payment of balance for phase 3 to suppliers that completed this phase satisfactorily

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Table 9. Detailed time schedule

2.9 IPR ISSUES

Ownership of results (foreground)

Each Supplier will keep ownership of the IPRs attached to the results they generate during the PCP implementation. The tendered price is expected to take this into account.

The ownership of the IPRs will be subject to the following:

- the Procuring Partners have the right to:
  - access results, on a royalty-free basis, for their own use
  - grant (or to require the suppliers to grant) non-exclusive licences to third parties to exploit the results under fair and reasonable conditions (without the right to sub-license)
- the Procuring Partners have the right to require the Suppliers to transfer ownership of the IPRs if the suppliers fail to comply with their obligation to commercially exploit the results *(see below)* or use the results to the detriment of the public interest *(including security interests)*.

Commercial exploitation of results

Commercial exploitation is an important part of a Pre-Commercial Procurement process. The Suppliers need to make a credible plan to secure access for the Procuring Partners to the solutions resulting from the R&D work done within FABULOS. It should be ensured that the Procuring Partners can continue to benefit from the solutions after the project has ended. Therefore, Suppliers are expected to protect their Intellectual Property and commercially exploit the results of the Research and Development undertaken in the PCP within a period of four years after the end of the Framework Agreement.

With respect to this, the Procuring Partners invite Suppliers to explore several innovative approaches and propose them with a future proof business model and commercialisation plan.

The business and commercialisation plan should explain the proposed approach to commercially exploit the results of the PCP and to bring a viable product or service to market. The business plan is expected to take into account the involved stakeholders and value chain. Also, an indicative investment- and exploitation plan as well as an indicative income plan (traveller fees and or public authority funding) for the pilot sites is expected. Based on the market analysis a first outlook should be made on the cost/benefit ratio in the transition towards full scale deployment. The plan should also list those topics that are very relevant for the commercialisation, but for which no quantitative data are available yet

Furthermore, a market analysis is meant to give a global qualitative insight in the market potential of the PCP services in both the pilot regions as well as on a European scale. The market analysis should at least indicate possible first-to-target deployment areas (geographical/services) and possible deployment barriers (e.g. political, technical, organisational, financial, ethical). Feasibility regarding principles for licensing, business models, pricing, and distribution should also be included.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Describe the innovation aspects of the proposed solution in respect to the state-of-the-art. This part should focus on the added value of the FABULOS solution compared to the current state-of-the-art on the market services (not compared to solutions still in pilot phases) with respect to a.o.:
- Comfort for users and impact on "willingness to pay"
- Cost-benefit ratio for transport operators
- Required public funding

A description of the expected size and type of the market in the six partner cities after the end of the PCP process can be found in **Tender Document 2 (functional specifications), Annex I**.

ISO standardisation for self-driving minibuses and other vehicles ("software driving license") is currently in its early stages of development. Suppliers must consider the future certification of their solutions or contribution to standardisation.

Suppliers need to foresee future requirements such as the implementation of "software driving license tests", that will likely been required by local authorities, for field testing the solutions during Phase 3.

The feasibility of the business plan to commercially exploit the R&D results (Form E, section 4) will be assessed as part of the award criteria. Furthermore, the commercialisation plan will be part of the End-of-Phase reports of all three Phases, as well as of the offers for the Phases 2 and 3.

In addition to the commercialisation activities performed by the suppliers, the FABULOS Procuring Partners will promote the R&D results via its network of prefered partners, which consists of several other public procurers and related organisations. The Procuring Partners will also actively disseminate the suppliers' results at the end of each phase via relevant public and industry related events. It is the goal of the Procuring Partners to help develop a working market for such type of solutions in order to ensure their usability and sustainability and to help to overcome possible, commonly defined deployment barriers.

One larger final event organised by the Procuring Partners is foreseen. During the event all end solutions will be presented and potential follow-up initiatives will be discussed at that time with the extended network of preferred partners.

Declaration of pre-existing rights (background)

The ownership of pre-existing rights will remain unchanged by the PCP.

In order to be able to distinguish clearly between results and pre-existing rights (and to establish which pre-existing rights are held by whom):

- Suppliers are requested to list the pre-existing rights for their proposed solution in their offers
- Procuring Partners and Suppliers will be requested to establish a list of pre-existing rights to be used before the start of the contract.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

The Procuring Partners do not hold any pre-existing rights relevant to the PCP contracts.

The framework agreement will contain a provision that describes in more detail the rights and obligations of the different parties regarding the pre-existing rights and results.

3. Evaluation of tenders

3.1 ELIGIBLE TENDERERS, JOINT TENDERS AND SUBCONTRACTING

Participation in the tendering procedure is **open** on equal terms to **all types of organisations from any country,** regardless of their geographic location, size or governance structure.

Tenders may be submitted by a **single entity** or in collaboration with others. The latter can involve either submitting a **joint tender** or subcontracting, or a combination of the 2 approaches.

Participation in the **open market consultation** is not a condition for submitting a tender.

Attention: There will, however, be a requirement relating to the place of performance of the R&D services *(see below)*.

For phases 2 and 3, participation is limited to Suppliers that successfully completed the preceding phase.

Where it is stated that Suppliers are to comply with administrative instructions, those that do not will be excluded from further participation in the FABULOS PCP.

3.1.1 Joint Tender or Tender submitted by a consortium

A Consortium (a combination of firms) may submit a joint Tender. Any type of natural or legal person (including non-profit entities properly registered like universities) shall be entitled to submit a Tender either individually or by way of an association or consortium comprising several Suppliers, set up temporarily for the purposes of the FABULOS PCP.

A joint Tender must specify the role, qualification and experience of each member of the consortium. A single authorized representative of the association or consortium, with sufficient powers to exercise the rights and comply with the obligations that arise from the FABULOS PCP procedure shall be appointed and be mandated as the Lead Tenderer (further named as Tenderer). The Lead Tenderer shall sign the Tender and the contracts in the name and on behalf of all members, and shall be responsible for all aspects and execution of the contracts without prejudice to the existence of joint powers that they may grant for receiving and making payments of a significant amount.

All members of the consortium shall be jointly and separately bound to fulfil the terms of the contracts. The Lead Tenderer shall be mandated to act on behalf of the consortium for the purposes of the contracts and shall have the authority to bind the consortium. The composition of the consortium shall not be altered without the prior consent of the Lead Procurer. Any alteration in the composition of the consortium without the prior consent of the Lead Procurer may result in the termination of the contracts.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

A Consortium statement should be signed by all suppliers agreed to set up a team to participate jointly in the FABULOS procedure, and to set up a temporary Consortium of Suppliers, to comply jointly with the purposes of the PCP procedure and with the contracts providing a statement from the supplier declaring that it is aware of the provisions set out in the Tender Documents (in particular in relation to IPRs).

Contact details of the Lead Tenderers must be stated in Form A. The names, circumstances and participation of the members of the association or consortium should be properly described.

3.1.2 Subcontracting

A subcontractor is a third party which has entered into an agreement on business conditions with one or more beneficiaries, in order to carry out part of the work of the project without the direct supervision of the beneficiary and without a relationship of subordination.

Subcontracting is permitted in each Phase of the FABULOS PCP procedure. No essential parts of the contracts can be subcontracted, nor the management of the PCP. According to Finnish legislation, the Act on the Contractor's Obligations and Liability when Work is Contracted Out (1233/2006) will be applied. This implies that the service provider must supply to the customer during the contract period, at twelve month intervals, a certificate of tax payment or a tax liability certificate, as well as a certificate on the taking out of pension insurance and the payment of pension insurance premiums.

The Supplier shall state in the Tender Submission Form (Form A) which part of the PCP obligations and contract performance, if any, is intended to be subcontracted to other Suppliers. The Supplier shall describe its approach in selecting and managing its subcontractors. Also in this form, the Supplier will identify who the subcontractor(s) is/are and which services they will deliver in the project. The Supplier shall provide a statement from the subcontractor declaring that it is aware of the provisions set out in the Tender Documents, that it meets the qualification requirements for the subcontracted service and that it has its resources at the Supplier's disposal for the full duration of the contract.

The Suppliers remain fully liable to the procurers for the performance of the contract. This is the reason why subcontracts must reflect the rules of the H2020 grant agreement, including as relates to the place of performance, the definition of R&D services, confidentiality, results and IPRs, the visibility of EU funding, conflicts of interest, language, obligation to provide information and keep records, audits and checks by the EU, the processing of personal data, liability for damages and ethics and security requirements.

3.1.3 Replacement of a subcontractor

If, subsequently, the Supplier needs to change or add new subcontractors (Phases 1 through 3), these new subcontractors must provide a statement with the same content described in the above paragraph and following the same form. Nevertheless, no change in subcontractor shall be possible if:

- It leads, according to an independent legal report, to IPR/confidentiality issues (i.e. if associated participants selected for Phase 1 decide to continue as subcontractor for another Supplier)
- It does not allow the Supplier maintaining the technical and financial capacity required

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Notwithstanding the grant of any subcontract, the Supplier remains responsible to the Procuring Partners for the performance and observance of all its obligations under the Framework Agreement and the Specific Contracts and for the consequences of any negligent acts of the subcontractors.

The execution of the tasks assigned to a subcontractor shall not be the subject of further subcontracting.

3.2 EVALUATION CRITERIA – OVERVIEW

The process to award the Framework Agreements and the Specific Contracts is based on four main categories:

- The **exclusion** criteria: evaluate the individual situation of a Tenderer;
- The **selection** criteria: determine whether a Tenderer has the financial, technical and professional capacity necessary to carry out and perform the work;
- The **compliance** criteria: evaluate if the submitted Tender is compliant with the principles of PCP, public financing, place of performance, research integrity and security;
- The **award** criteria: award contracts to the best ranked Tenders.

3.3 EXCLUSION CRITERIA

The purpose of the exclusion criteria is to determine the situation of the Suppliers and subcontractors. The situation of the economic operator will be assessed based on responses to questions in Form B on a pass/fail basis.

A Supplier will be excluded from further participation in the FABULOS PCP if it, or any subcontractor on whose resources it relies upon in this procurement, does not meet one or several of the exclusion criteria.

The exclusion criteria are as follows:

Exclusion criteria	Required Evidence
A) Conflict of Interest	A) a declaration of honour stating the absence of Conflict of Interest, Bankruptcy and professional misconduct or Criminal offences (template)
B) Criminal offences	
C) Bankruptcy and professional misconduct	

Table 10. Exclusion criteria

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Suppliers must confirm, by signing a declaration of honour, that they are not subject to any of the exclusion criteria listed below.

Suppliers that do not comply with these criteria will be excluded.

A) Conflict of interest

Suppliers that are subject to a conflict of interest may be excluded. If there is a potential conflict of interest, Suppliers must immediately notify the lead procurer in writing.

A conflict of interest covers both personal and professional conflicts.

Personal conflicts are any situation where the impartial and objective evaluation of tenders and/or implementation of the contract is compromised for reasons relating to economic interests, political or national affinity, family, personal life (e.g. family of emotional ties) or any other shared interest.

Professional conflicts are any situation in which the Supplier's (previous or ongoing) professional activities affect the impartial and objective evaluation of tenders and/or implementation of the contract.

If an actual or potential conflict of interest arises at a later stage (i.e. during the implementation of the contract), the Supplier must contact the lead procurer, who is required to notify the EU and to take steps to rectify the situation. The EU may verify the measures taken and require additional information to be provided and/or further measures to be taken.

Suppliers shall - for each of the PCP Phases - explicitly confirm that they are not subject to any of the exclusion criteria listed above and shall sign a declaration of honour stating the 'absence of a conflict of interest'.

See Declaration confirming the absence of any conflict of interest in Form B, Part B.

B) Criminal offences

Suppliers must confirm, by signing a declaration of honour, that they are not subject to any of the exclusion criteria listed below, see Form B, Part A1:

- Criminal offences referred to in Article 2 of Council Framework Decision 2008/841/JHA of 24 October 2008 on combating organized crime;
- Corruption as defined in Article 3 of Council Act of 26 May 1997 preparation on the basis of Article K.3.2 c Treaty on European Union, the Convention on the fight against corruption involving officials of the European Communities or officials of Member States of, and Article 3.1 Council Joint Action 98/742/JHA of 22 December 1998 adopted by the Council on the basis of Article K.3 of the Treaty on European Union, on corruption in the private sector;
- Fraud within the meaning of Article 1 of the Convention drawn up on the basis of Article K.3 of the Treaty on European Union for the Protection of the Communities' financial interests;

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

- Money laundering as defined in Article 1 of Council Directive 91/308/EEC of 10 June 1991 on measures to prevent the financial system for money laundering, amended by European Parliament and Council Directive 2001/97/EC;
- Terrorist offences or offences linked to terrorist activities as defined in Articles 1 and 3 of Council Framework Decision of 13 June 2002 on combating terrorism;
- Child labour and other forms of trafficking in human beings as defined in Article 2 of Directive 2011/36/EU of the European Parliament and of the Council of 5 April 2011 on preventing and combating trafficking in human beings and protecting its victims, and replacing Council Framework Decision 2002/629/JHA;
- Is guilty of serious misrepresentation in supplying the information required under this Section or has not supplied such information.

If the Procuring Partners become aware that a Supplier or a representative of the Supplier or subcontractor, under a judgment that has entered into final legal force has been sentenced for a criminal offence listed above, such Supplier or subcontractor, will be excluded from the FABULOS PCP.

C) Bankruptcy and professional misconduct

A Supplier will be excluded from participation if he:

- Is bankrupt or is being wound up, is under compulsory administration or is the subject of a composition or has indefinitely stopped its payments or is subject to a prohibition on conducting business;
- Is the subject of proceedings for a declaration of bankruptcy, for an order for compulsory winding up or administration by the court or composition or any other similar proceedings;
- Has been convicted by a judgment which has the force of res judicata for an offence relating to professional practice;
- Has been guilty of grave professional misconduct and the procuring agencies can prove this;
- Has not fulfilled its obligations relating to social insurance charges or tax in its own country;
- In some material respect has failed to provide information requested or provided incorrect information required pursuant to this invitation to tender document.

3.4 SELECTION CRITERIA

The purpose of the selection criteria is to determine whether a Supplier has the financial, economic, technical and professional capacity necessary to carry out and perform the work. These selection criteria will be evaluated on a pass/fail basis.

“Fail” means that the evidence given does not provide sufficient indication of the Supplier’s expertise, ability and/or equipment to meet project’s objectives. Any Supplier that cannot meet all requirements in this Section will not be selected.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

The selection criteria are as follows:

Ability to perform R&D up to original development of the first products or services and to commercially exploit the results of the PCP, including intangible results in particular IPRs

Suppliers must have:

- the capacity, tools, material and equipment to:
  - carry out research and lab prototyping
  - produce and supply a limited set of first products or services and demonstrate that these products or services are suitable for production or supply in quantity and to quality standards defined by the procurers
- the financial and organisational structures to
  - manage, exploit and transfer or sell the results of the PCP *(including tangible and intangible results, such as new product designs and IPRs)*
  - generate revenue by marketing commercial applications of the results *(directly or through subcontractors or licensees)*.

To do so, he is asked to provide at the following evidence:

Provide **a description of relevant reference and/or previous projects** which reflect the competences and capacity of the Tenderer in the different phases and domains of the FABULOS project (cfr. research, development, prototyping, testing and commercialisation).

These references can be provided based on previous projects of the Supplier or one or several of the consortium partners and/or subcontractors who will be working on the project. These projects should be completed in the last 5 years. Examples of areas in which references can be provided: large scale data projects, integration projects with sensors and other devices, experience with privacy and security in a network context ...

In describing these reference projects Suppliers will provide:

- Proof of the capacity, tools, material and equipment to carry out research, development, testing and lab prototyping, as well as proof of the capacity to produce and supply a limited set of first products or services, and demonstrate that these products or services are suitable for production or supply in quantity and to quality standards defined by the procurers (cfr. a prototype)

- Proof he is able to manage, exploit and transfer or sell the results of the PCP (including tangible and intangible results, such as new product designs and IPRs) and generate revenue by marketing commercial applications of the results (directly or through subcontractors or licensees)

- Proof he is able to provide the necessary competences to complete this project.

Therefore, each Supplier will provide a number of **CVs of key personnel and competences** which he deems necessary to complete the project..

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Confirm that the Supplier has a Business Continuity / Disaster Recovery / Risk Management plan that ensures that the described services are delivered in the event of a disruption affecting its business, and ensures continuity of supply / service from its critical suppliers.

Confirm whether the Supplier will take the appropriate level of insurance cover if he is to be successful in winning the contract.

The Supplier needs to keep in mind that the submitted CVs and references will have to showcase their abilities in all domains and phases of the FABULOS project.

Each Supplier shall - **for each of the PCP Phases** - describe, present and confirm the required references and competences in Form C. Should there be any doubt as to any of these criteria, the Supplier may be requested to provide additional information.

3.5 COMPLIANCE CRITERIA

There are 2 types of award criteria: compliance criteria (or: on/off criteria) and weighted criteria.

Compliance criteria (or: On/off award criteria)

The purpose of the compliance criteria is to determine whether the Tender is compliant with the principles of PCP, public financing, place of performance, research integrity and security.

These compliance criteria will be evaluated on a pass/fail basis, based on the responses to the questions in Form D. The offers for each phase will be evaluated against these criteria.

Suppliers and their Tenders must comply with all of the following on/off award or compliance criteria (this also applies to the call-off for phases 2 and 3):

Compliance criteria	Evidence
A) Compliance with the definition of R&D services	See explanation below
B) Compatibility with other public financing	See explanation below
C) Compliance with the requirements regarding the place of performance of the contract	See explanation below
D) Compliance with ethics requirements	See explanation below
E) Compliance with security requirements	See explanation below

Table 11. Compliance criteria

Tenders that do not comply with these criteria will be excluded.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

A) Compliance with the definition of R&D services

Tenders that go beyond the provision of R&D services will be excluded.

R&D covers fundamental research, industrial research and experimental development, as per the definition given in the EU R&D&I state aid framework[8]. It may include exploration and design of solutions and prototyping up to the original development of a limited volume of first products or services in the form of a test series. Original development of a first product or service may include limited production or supply in order to incorporate the results of field-testing and to demonstrate that the product or service is suitable for production or supply in quantity to acceptable quality standards.[9] R&D does not include quantity production or supply to establish commercial viability or to recover R&D costs. It also excludes commercial development activities such as incremental adaptations or routine or periodic changes to existing products, services, production lines, processes or other operations in progress, even if such changes may constitute improvements. The purchase of commercial volumes of products or services is not permitted.

The definition of services means that the value of the total amount of products covered by the contract must be less than 50 % of the total value of the PCP framework agreement.

The following evidence is required:

- the financial part of the offer for the framework agreement must provide binding unit prices for all foreseeable items for the duration of the whole framework agreement
- the financial part of the offer for each phase must give a breakdown of the price for that phase in terms of units and unit prices for every type of item in the contract, distinguishing clearly the units and unit prices for items that concern products
- the offers for all 3 phases may include only items needed to address the challenge in question and to deliver the R&D services described in the request for tenders
- the offers for all 3 phases must offer services matching the R&D definition above
- the total value of products offered in phase 1 respectively phase 2 must be less than 50 % of the value of the phase 1 respectively phase 2 contract and the total value of products offered in phase 3 must be so that the total value of products offered in all phases (1,2 and 3) is less than 50% of the total value of the PCP framework agreement.

Suppliers shall - for each of the PCP Phases - provide a financial offer. See Form F.

B) Compatibility with other public financing

Tenders that receive public funding from other sources will be excluded if this leads to double public financing or an accumulation of different types of public financing that is not permitted by EU legislation, *including EU state aid rules*.

Suppliers shall - for each of the PCP Phases - sign a declaration of honour stating the 'absence of

[8] See Point 15 of the Commission Communication on a framework for state aid for research and development and innovation (C(2014) 3282).

[9] See Article XV(1)(e) WTO GPA 1994 and the Article XIII(1)(f) of the revised WTO GPA 2014.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

other incompatible public financing'. See Form D.

C) Compliance with requirements relating to the place of performance of the contract

Tenders will be excluded if they do not meet the following requirements relating to the place of performance of the contract:

- at least 50% of the total value of activities covered by each specific contract for PCP phase 1 and 2 must be performed in the EU Member States or in H2020 associated countries. The principal R&D staff working on each specific contract must be located in the EU Member States or H2020 associated countries.

- at least 50% of the total value of activities covered by the framework agreement *(i.e. the total value of the activities covered by phase 1 + the total value of the activities covered by phase 2 + the total value of the activities covered by phase 3)* must be performed in the EU Member States or H2020 associated countries. The principal R&D staff working on the PCP must be located in the EU Member States or H2020 associated countries.

This 50% rule applies per Supplier. If suppliers apply as a consortium, the 50% rules applies thus to the consortium as a whole, not to each individual supplier within a consortium.

The percentage is calculated as the part of the total monetary value of the contract that is allocated to activities performed in the EU Member States or in other countries associated to Horizon 2020. All activities covered by the contract are included in the calculation *(i.e. all R&D and operational activities that are needed to perform the R&D services, e.g. research, development, testing and certifying solutions)*. This includes all activities performed under the contract by suppliers and, if applicable, their subcontractors.

Suppliers with suppliers from the UK need to clarify how they intend to keep on meeting the above-mentioned requirement in the case that the UK does not become an associated country to the EU after Brexit.

The principal R&D staff are the main researchers, developers and testers responsible for leading the R&D activities covered by the contract.

The countries associated to Horizon 2020 are those listed as associated countries in the Participant Portal Online Manual[10].

The following evidence is required:

- the financial part of the offer must provide binding unit prices for all foreseeable items for the duration of the whole framework agreement and give a breakdown of the price for the current phase in terms of units and unit prices *(hours and unit price per hour)*, for every type of item in the contract *(e.g. junior and senior researchers)*

10 List of H2020 associated countries.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

- a list of staff working on the specific contract *(including for subcontractors)*, indicating clearly their role in performing the contract *(i.e. whether they are principal R&D staff or not)* and the location *(country)* where they will carry out their tasks under the contract
- a confirmation or declaration of honour that, where certain activities forming part of the contract are subcontracted, subcontractors will be required to comply with the place of performance obligation to ensure that the minimum percentage of the total amount of activities that has to be performed in the EU Member States or in countries participating in Horizon 2020 is respected'

Suppliers shall - for each of the PCP Phases - provide a financial offer. See Form F.

D) Ethics and research integrity

Tenders will be excluded if they:

- do not comply with the following rules:
  - ethical principles *(including the highest standards of research integrity, notably as set out in the European Code of Conduct for Research Integrity[11], and, in particular, avoiding fabrication, falsification, plagiarism and other research misconduct)*
  - applicable international, EU and national law
- include plans to carry out activities in a country outside the EU if they are prohibited in all Member States
- include activities that do not focus exclusively on civil applications

In Phase 3, tendered will need to comply, if applicable, with the following ethics requirements:

- give details on the procedures and criteria that will be used to identify/recruit research participants must be provided.
- Detailed information must be provided on the informed consent procedures that will be implemented for the participation of humans. The applicant should also detail measures to ensure that, in the case of company employees taking part as participants in testing, their consent is fully voluntary and informed, and not unduly influenced by concerns for their future employment.
- Templates of the informed consent forms and information sheet must be submitted.
- The applicant must clarify whether children and/or adults unable to give informed consent will be involved and, if so, justification for their participation must be provided as well as clarification how consent/ assent will be ensured.
- The applicant must clarify whether vulnerable individuals/groups will be involved. Details must be provided about the measures taken to prevent the risk of enhancing vulnerability/stigmatisation of individuals/groups.
- Copies of ethics approvals for the research with humans of the relevant Ethics Committee or Body must be submitted on request.
- Copies of opinion or confirmation by the competent Institutional Data Protection Officer and/or authorization or notification by the National Data Protection Authority must be

11 The European Code of Conduct for Research Integrity of ALLEA (All European Academies).

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

submitted (whichever applies according to the EU Data Protection Laws (EC Directive 95/46, Regulation (EU) 2016/679, Directive (EU) 2016/680) and the national law).

- Detailed information must be provided on the procedures that will be implemented for data collection, storage, protection, retention and destruction and confirmation that they comply with national and EU legislation. (including the new data protection laws that came into force in May 2018 (Regulation (EU) 2016/679 and Directive (EU) 2016/680).
- Templates of the informed consent forms and information sheet must be submitted.
- The applicant must ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/ legislation are followed for staff involved in this project and provide detailed descriptions on the safety protocols and methods (incl. technical) to be applied during the testing activities.
- Respective certificates/ authorisations should be provided, where relevant, for any dual-use items used in the final solution.
- A risk assessment with elaboration on the applicable legal requirements and measures to prevent misuse must be provided.
- The applicants should give details of how they will address the possible social and economic impacts of the research and its results on future employment patterns for workers in the sector and what measures could be proposed to mitigate possible loss of jobs for them. Such considerations can be included in works within task T5.3 Policy roadmap and procurement guide for future uptake of FABULOS results and be summarised as a part of deliverable D5.6: Policy paper. They can also be incorporated in the management/ dissemination and communication activities of the project.

Before starting the particular task that raises ethical issues, Suppliers must provide a copy of any ethics committee opinion required under national law; and any notification or authorisation for activities raising ethical issues required under national law. In addition, if the tender involves activities that raise ethical issues, the Supplier must submit an ethics self-assessment (see the guidance for EU grant beneficiaries How to complete your ethics self-assessment).

Call-offs for phases 2 and 3 may request that this information be updated in the offers submitted for these phases.

E) Security

Tenders will be excluded if they do not comply with EU, national and international law on dual-use goods or dangerous materials and substances.

Tenders and the results of the executed work must not contain any classified information.

If the output of activities or results proposed in the tender raise security issues or uses EU-classified information, the Supplier must show that these issues are being handled correctly. In such a case, Supplier are required to ensure and to provide evidence of the adequate clearance of all relevant facilities. They must examine any issues (*such as those relating to access to classified information or export or transfer control*) with the national authorities before

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

submitting their offer. Tenders must include a draft security classification guide (SCG), indicating the expected levels of security classification.

If necessary for the tender procedure or for performing the contract itself, Suppliers will be requested to ensure appropriate security clearance for third parties *(e.g. for personnel)*.

Call-offs for phases 2 and 3 may request that this security information be updated in the offers submitted for that phase.

Before starting the particular task that raises security issues, Suppliers must provide a copy of any export or transfer licences required under EU, national or international law.

The framework agreement and/or the specific contracts contain a provision on security.

For information on security, see the guidance for EU grant beneficiaries: Guidelines for the handling of classified information in EU research projects.

Should there be any doubt as to any of these criteria, Suppliers may be requested to provide additional information.

3.6 AWARD CRITERIA

Weighted award criteria

The award will be made based on the Most Economically Advantageous Tender (MEAT), based on the best quality price ratio.

In Tender Document 2, the Functional and non-functional Specifications are described in chapters 3 and 4. These requirements are expected to be a part of the end-solution, by the time the FABULOS project finishes. The FABULOS partners do not expect Suppliers to already have all these features in place when submitting their tender; this work is part of the R&D process. In the Technical Offer (via Form E), Suppliers need to make clear how their intend to achieve the must haves and (if and) how they will implement the nice to haves. These explanations will be appraised by the Technical Evaluation Committee, assisted by a panel of external experts. The Technical offer will not be evaluated on a pass/fail basis: failure to (sufficiently) describe the "must haves" is therefore not a reason to be excluded from this tender, but will merely lead to lower evaluation scores.

The evaluation will be assessed based on the following criteria. The model Appendix 4 "-Scoring Model for the Award Criteria" will be used to assess and score the extent to which a Tender meets the award criteria.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Weighted award criteria	Maximum points	Thresholds	Weighting
Award criteria Phase 1 and Framework Agreement			
Functional criteria			**35%**
FR1): Fleet Management Integration system	**20**	**10**	
FR2) Control Room functions and remote operation	**15**	**7**	
FR3 & 5): Integration in city City traffic control systems, public transport systems and traffic infrastructure	**20**	**10**	
FR4): Maintenance and incident management	**10**	**5**	
FR6): Traffic situation capabilities	**10**	**5**	
FR7&8): Vehicle, fleet and operational requirements	**20**	**10**	
FR9): Deployment, set-up and service	**5**	**2**	
Total	***100***		
Project management criteria			**20%**
PM1) Feasibility of the Project plan and schedule, including methodology, risk management and quality assurance	**70**	**35**	
PM2) Composition of the project team	**30**	**15**	
Total	***100***		
Non-functional criteria			**30%**
NFR1) Safety and technical maturity	**50**	**25**	
NFR2) Societal Maturity	**25**	**12**	
NFR3) Legal Maturity	**25**	**12**	
Total	***100***		
Commercial feasibility			**5%**
CF1) Completeness, sense of reality and feasibility of the commercialisation plan including the market analysis and risk management. Sense of reality and feasibility of the principles for licensing, pricing, distribution	**100**	**50**	
Total	***100***		
Price			**10%**
P1) Binding contract price for carrying out the work in the present phase	**25**		
P2) Indicative price for the work in all three PCP Phases, incl. the present phase (total price)	**75**		
Total	***100***		

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Weighted award criteria	Maximum points	Thresholds	Weighting
Award criteria Phase 2: Prototyping and lab testing			
Functional criteria			30%
FR1): Fleet Management Integration system	15	8	
FR2) Control Room functions and remote operation	15	8	
FR3 & 5): Integration in city City traffic control systems, public transport systems and traffic infrastructure	20	10	
FR4): Maintenance and incident management	10	5	
FR6): Traffic situation capabilities	10	5	
FR7 & 8): Vehicle, fleet and operational requirements	20	10	
FR9): Deployment, set-up and service	10	5	
Total	***100***		
Project management criteria			15%
PM1) Feasibility of the Project plan and schedule, including methodology, risk management and quality assurance	70	35	
PM2) Composition of the project team	30	15	
Total	***100***		
Non-functional criteria			30%
NFR1) Safety	50	25	
NFR2) Societal Maturity	25	12	
NFR3) Legal Maturity	25	12	
Total	***100***		
Commercial feasibility			10%
CF1) Completeness, sense of reality and feasibility of the commercialisation plan including the market analysis and risk management. Sense of reality and feasibility of the principles for licensing, pricing, distribution	100	50	
Total	***100***		
Price			15%
P1) Binding contract price for carrying out the work in the present phase	25		

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

P2) Indicative price for the work in remaining PCP Phases, incl. the present phase (total price)	**75**		
Total	***100***		

Weighted award criteria	**Maximum points**	**Thresholds**	**Weighting**
Award criteria Phase 3: Field testing			
Functional criteria			**35%**
FR1): Fleet Management Integration system	**15**	**8**	
FR2) Control Room functions and remote operation	**15**	**8**	
FR3 & 5): Integration in city City traffic control systems, public transport systems and traffic infrastructure	**15**	**8**	
FR4): Maintenance and incident management	**10**	**5**	
FR6): Traffic situation capabilities	**10**	**5**	
FR7 & 8): Vehicle, fleet and operational requirements	**20**	**10**	
FR9): Deployment, set-up and service	**15**	**8**	
Total	***100***		
Project management criteria			**15%**
PM1) Feasibility of the Project plan and schedule, including methodology, risk management and quality assurance	**65**	**35**	
PM2) Composition of the project team, incl years of relevant experience of key personnel	**20**	**10**	
PM3) Qualifications for operators	**15**	**8**	
Total	***100***		
Non-functional criteria			**20%**
NFR1) Safety	**50**	**25**	
NFR2) Societal Maturity	**25**	**12**	
NFR3) Legal Maturity	**25**	**12**	
Total	***100***		
Commercial feasibility			**20%**
CF1) Completeness, sense of reality and feasibility of the commercialisation plan including the market analysis and	**100**	**50**	

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

risk management. Sense of reality and feasibility of the principles for licensing, pricing, distribution			
Total	***100***		
Price			**10%**
P1) Binding contract price for carrying out the work in the present phase	**100**		
Total	***100***		

Table 12. Weighted Award criteria

See Tender Document 2: Functional Specifications for more details on each weighted award criterion.

Calculation example:

As per Appendix 4, Suppliers can receive the following scores for each of the criteria mentioned above:
0: Nonexistent
1: Very weak
2: Weak
3: Good
4: Very good
5: Excellent

If the score is “good” (i.e. 3) on FR1, the supplier receives 3 (score) x 20 (max. points) / 5 (max score) = 12 points. This is above the 10-point threshold. The same approach is applied to FR1-FR9. Then all scores are added, leading to a total number of points for Functional Criteria. This is multiplied by the weighting percentage - in this case 35% - leading to a final score for that section. All sections are dealt with in this same way and all scores per section are added to come to one final end score.

The criteria list for phases 2 and 3 may be subject to change. Additional sub-criteria may be added for the call-offs for phases 2 and 3, as a way of making the award criteria more precise, provided that they do not substantially change the existing criteria. The final list of criteria for these phases will be provided with the call-off documentation. Weight scores per selection criterion for Phases 2 and 3 are also subject to confirmation in the call-off documents.

Should there be any doubt as to any of these criteria, Suppliers may be requested to provide additional information.

Note:
The weighted award criteria must ensure that the procurer gets the best value for money. It is therefore not permitted to use either lowest price as the sole criteria, without taking quality into account, or highest quality as the sole criteria, without taking price into account.

All the award criteria will be evaluated by examining the written tender.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

3.7 EVALUATION PROCEDURE: OPENING OF TENDERS & EVALUATION

Opening of tenders

Tender submission takes places electronically, via the FABULOS website. The Tenders will be opened on 1 November 2018 at 09:00 CET. Tenderers cannot be present.

For the opening of the Tenders, the Lead Procurer will appoint an Administrative Evaluation Committee composed of two representatives of Forum Virium Helsinki.

The Administrative Evaluation Committee will be in charge of opening the Tenders and checking their general administrative compliance with the conditions on the content and format of the Tender.

The Lead Procurer will receive the proposals filed before the corresponding deadline in each Phase of the FABULOS PCP Procedure, opening them in the term described in this Request for Tender, as well as in the Specific Contract call-offs.

A report is compiled of this opening session. This report contains all the information about the opening. All submitted Tenders are automatically included in the report. The report is then signed by at least two representatives of the Administrative Evaluation Committee.

Tenders not complying with the formal and procedural requirements will be excluded from the Tender evaluation.

Evaluation

The Technical Evaluation committee will evaluate the tenders, carrying out the following four steps:

- Step 1 — Checking whether the exclusion criteria apply to the tenderer (pass/fail; based on Form B)
- Step 2 — For tenderers passing Step 1, evaluating the tender based on the compliance (or: on/off award) criteria (pass/fail; based on Form D)
- Step 3 — For tenderers passing Step 2, assessing whether the tenderer has the capacities necessary to perform the contract, on the basis of the selection criteria (based on Form C)
- Step 4 — For tenders passing Step 3, evaluating the tender based on the weighted award criteria (Technical Offer, Form E)

For the evaluation of the Tenders, the Lead Procurer will appoint ad personam the Technical Evaluation Committee, composed of an odd number of maximum 7 people for the evaluation of the Tenders based on the award criteria. The Technical Evaluation Committee will be composed by representatives of all FABULOS project partners in an equal and balanced way.

Tenders to the PCP Request for Tender, as well as the Tenders for the Phase 2 and 3 call-offs, will preferably be assessed and ranked by the same Technical Evaluation Committee. The FABULOS Procuring Partners hold the right to replace evaluators during the project provided that the replacement evaluator has the necessary skills and represents the same member of the

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Procuring Partners.

The Procuring Partners have the right to ask external experts for support. Therefore, in addition to the Technical Advisory Committee, an Advisory Evaluation Panel will be appointed. This panel will consist of external stakeholders with specific expertise on (elements of) the FABULOS challenge. Members of the Advisory Evaluation Panel will for example have special expertise in road safety, cyber security, public transport and business case development.

The Advisory Evaluation Panel will be provided with all the eligible Technical Offers. They are asked to read (at least) the sections related to their expertise. In a (physical) workshop with the Technical Evaluation Committee, each expert presents an advice and a ranking of the offers. The Technical Evaluation Committee has an obligation to re-evaluate the Tenders, if the advice of the Advisory Evaluation Panel is contradictory to the ranking of the Technical Evaluation Committee.

The evaluation process will be conducted as follows:

First of all, member of the Technical Evaluation Committee will independently and individually assess all Tenders. The more points a Tender scores in total, the higher it is ranked. Based on the evaluators' individual assessments, which are all equally weighted, a preliminary ranking of the Tenders will be made.

If deemed necessary, an online hearing will take place, where Suppliers will be asked to clarify aspects of their Tender.

Subsequently, the Technical Evaluation Committee will meet to collect, compare and discuss the comments of each evaluator for each Tender, and to review the preliminary scoring and ranking to ensure that the assessment of all Tenders is consistent and non-discriminatory.

When this is done, the Advisory Evaluation Panel will present its advice and a ranking of the offers to the Technical Evaluation Committee. The Technical Evaluation Committee has an obligation to re-evaluate the Tenders, if the advice of the Advisory Evaluation Panel is contradictory to the ranking of the Technical Evaluation Committee.

By consensus or, if that fails, by a majority of two thirds, the Technical Evaluation Committee will then make the final recommendations for the award of the contracts to the Procurers Steering Committee (PSC). The PSC will ultimately decide on the final ranking of the Tenders and the award of the Contracts.

An assessment of the Suppliers' capacity to perform the contract will be done on the basis of the selection criteria. The Suppliers shall be able to satisfy all of the criteria. These will be evaluated on a pass/fail (yes/no) basis.

Only Tenders passing this step will be further evaluated by the Technical Evaluation Committee. The compliance criteria will be assessed on the basis of the declaration made in Form D, followed by the assessment of the Technical Offer (Form E). The evaluation of the Technical Offer will be done based on technical and non-technical award criteria and according to the scoring model (see Appendix 4 - Scoring Model for the Award Criteria to this Request for Tenders). A minimum threshold of 60% of the total amount of points (excluding the price) must be reached.

The financial Section of Tenders passing this step will then be evaluated. The Financial Offer

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

(Form F) will be assessed on the basis of the formula foreseen in the scoring model (Appendix 5 - Scoring Model for the Price of the Request for Tenders). The price that will be evaluated is the Actual Price.

Large differences in assessment by the evaluators will be identified. If the reasoning given by the evaluators requires further clarification, this is provided by them.

The final award will be made on the basis of the Most Economically Advantageous Tender (MEAT). The expected minimum number of awarded contracts for Phase 1 is ten.

After the award decision has been taken, the Suppliers will be informed about their ranking. To the selected Suppliers a Framework Contract and Specific Contract for Phase 1 will be sent shortly thereafter. Suppliers will have approximately 2 weeks to sign the contracts.

Successful Suppliers will be, if awarded a contract, expected to start and finish their project in time. If the project is not finished by the deadline, the supplier will not be eligible to submit a Tender for the next Phase.

Unsuccessful Suppliers, that have not been excluded and who comply with the selection criteria, may contact the Lead Procurer, within 10 days, to obtain additional information about their Tender not being selected for a contract. Suppliers will be given feedback on their tender.

Evaluation of Phases 2 and 3

The criteria and the method for evaluating the tenders in Phase 2 and 3 will be the same as the criteria and the method used in evaluating the original tenders as set out in this chapter, but may be elaborated or developed in further detail within those frames. The weighting of each award criterion may differ from the initial weight in Phase 1 or Phase 2.

For Phase 2 and Phase 3, the composition of the Evaluation Committee and evaluation process up to the award decision will, as much as possible, remain the same as for Phase 1. Nonetheless, the evaluation process may be described in more detail, in particular for Phase 3, in respect to the field testing procedures.

Assessment and scoring

Only tenders with the following minimum scores (threshold) are eligible for consideration for a contract:

- 60% of the maximum number of points for each of the criteria, excluding Price:
  - Project Management
  - Technical quality of the solution (Functional requirements)
  - Non-functional requirements
  - Commercial feasibility

- 60% of the maximum number of points for the combined scores, including Price.

Failure to achieve the minimum score for any of the Tender components will result in the Tender being excluded from further participation in the PCP.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

The assessment criteria, weighting and the maximum points available are listed in paragraph 3.6. The Scoring Model is found in the Appendix 4 - Scoring Model for the Award Criteria.

The scoring will be made according to an absolute scale, meaning that several Tenders can receive the same score and that the score a particular Tender receives, is not affected by the scores other Tenders have received.

Communication and Q&A
Two webinars will be organised (on 11.9.2018 and 15.10.2018) to clarify the Call for Tender Documents, the procedure and to answer potential Suppliers' questions or requests for clarification.

All questions or requests for clarification must be received by Forum Virium Helsinki no later than 16 October 2018 at 17:00 CET. Any questions received after this deadline will not be answered.

The questions or requests for clarification must be addressed by email to Ask@fabulos.eu
With each question the correct document reference (Tender Document 1; Form E…) and page number should be clearly stated.

All questions and answers will be presented in an anonymised Q&A document that will be published on fabulos.eu in English (final version planned for 21.10.2018).

For phases 2 and 3, the Q&A will not be published, but only distributed to all suppliers that successfully completed the previous phase.

Unless otherwise instructed, please do not use any other contact addresses or means or contact any other persons in connection with this procurement.

4. Content & format of tenders

4.1 FORMAT

Tenders shall be received at no later than the closing date **31 October 2018 17:00 PM CET.**

All Tenderers must use the FABULOS Tender forms, which can be accessed along with all of the other Tender Documents by following the instructions in the Contract Notice on TED. The Tender documents are published on the FABULOS Website and can be downloaded from there.

All Tenders must be submitted as follows:

1. Tenders **have to be submitted electronically** via the fabulos.eu website (request for tenders page)
2. Tenders **shall contain an administrative, a technical and a financial section**, see Forms A through G;

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

3. The Tender, i.e. the Tender Submission Forms (Forms A through G) and all attachments, mandatory or not, will be **signed by the Supplier**, electronically or in "blue ink".

More information on the electronic submitting of the Tenders in Appendix 3– Electronic submission of the FABULOS Tender.

Each Supplier carries the sole responsibility for the accurate, timely and complete uploading of its unique and only tender. Tenders which are not compliant to the above mentioned conditions will be regarded as irregular and will not be retained.

Only one Tender from a Supplier as main contractor will be accepted. Please do not submit the Tender on paper or submit more than one electronic Tender. The submission of a backup copy in any form is not allowed.

Tenders must be submitted in PDF format. Visuals can be added in attachment at JPG or PNG. Attached publications like brochures and promotional material are allowed, but will not be taken into account as part of the evaluation.

If the Tender exceeds a page limit then all words and/or pages in excess of the specified limit may not be considered further. Suppliers will use a minimum font size of 10. For the table with page limits per Annex (Tender Submission Forms), see Appendix 7 - Table of page limits

More specific information about the requirements for the Phase 2 and 3 Tenders will be provided in the Phase 2 and 3 call-offs.

Where it is stated that Suppliers are to comply with the administrative instructions, those that do not comply will be excluded from further participation in the Tender procedure. Tenders that do not comply with the selection and compliance criteria will automatically be rejected. The Lead Procurer's decision as to whether or not a Tender complies with these instructions will be final.

4.2 ADMINISTRATIVE SECTION

In order to be eligible, Suppliers shall submit the following documents and declarations as listed in the indicated order below:

Form A - General Tender Submission Form
Form B - Exclusion Criteria (declaration)
Form C - Selection Criteria
Form D - Compliance Criteria (declaration)
Form E - Technical Offer
Form F - Financial Offer and Cost Breakdown
Form G - Financial Offer Phase 1

The Supplier is by its Tender bound by a validity period of 120 calendar days, starting from ultimate deadline for submission, i.e. 31 October 2018.

The Lead Procurer may request clarification or additional evidence or amplification of details provided. In accordance with the principle of equal treatment, no alterations to Tenders are to be

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

sought or accepted through requests for clarifications. In case the provided clarification is found not compliant with what was requested, the Tender will be excluded from further evaluation.

Responses to the questions in the Forms B (Exclusion Criteria) and C (Selection Criteria) will be assessed as pass/fail. Only Tenderers achieving a “pass” for all criteria will be put forward for further evaluation.

More detailed information for the Phase 2 and 3 offers will be provided in the Phase 2 and 3 call-offs.

4.3 TECHNICAL SECTION

Tenders must include a detailed Technical Offer for Phase 1, containing:

- A technical plan that outlines the Tenderer's idea for addressing all the requirements given in the PCP challenge description, relating both to functionality and performance; and the technical details of how this would be implemented. This technical plan must include an explanation of the method, a work plan including time schedule, deliverables and milestones as detailed in the Request for Tenders and the Functional Specification. The Tender must specify the plans and objectives of the subsequent Phases 2 and 3 and beyond.
- A draft business plan that explains the proposed approach to commercially exploit the results of the PCP and to bring a viable product or service onto the market.
- A risk assessment and risk mitigation strategy.
- A reply to the question "Does this tender involve ethical issues? (YES/NO)" and if YES, an ethics self-assessment, with explanations how the ethical issues will be addressed.
- A reply to the question "Does this tender involve: activities or results that may raise security issues and/or EU-classified information as background or results?
- A list of the pre-existing rights (Background) relevant to the Tenderer's proposed solution, in order to allow IPR dependencies to be assessed within 30 days following the awarding of the Framework Agreement, see PCP time schedule in PCP Time Schedule.

Tenders failing to meet these requirements will be excluded.

The information provided in the technical Sections of the tender will be used to evaluate the Tenders, on the basis of the technical award criteria and the compliance criteria A (Compliance with the definition of R&D services), D (Ethics and research integrity) and E (Security).

More detailed information for the Phase 2 and 3 offers (in particular on the technical implementation plan, the updated business plan and the list of IPRs) will be provided in the Call-offs.

Form E (technical offer template) should be used for drafting the technical section of the tender.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

4.4 FINANCIAL SECTION

The tender must include a detailed **financial offer** specifying:

- binding **unit prices** for all items needed for carrying out phase 1 and for items that are expected to be needed for phases 2 and 3 *(given in euros, excluding VAT but including any other taxes and duties)*
- a fixed **total price** for phase 1 and an estimated total price for phases 2 and 3, broken down to show unit prices and the number of each unit needed to carry out phase 1 *(given in euros, excluding VAT but including any other taxes and duties)*.

- In addition, the financial section must include:
- a **price breakdown** that shows the price for R&D services and the price for supplies of products (to demonstrate compliance with the definition of R&D in on/off award criterion A)
- a **price breakdown** that shows the location or country in which the different categories of activities are to be carried out *(e.g. x hours of senior researchers in country L at y euro/hour; a hours of junior developers in country M at b euro/hour)* (to demonstrate compliance with the requirement relating to place of performance in on/off award criterion C)
- the **financial compensation** valuing the benefits and risks of the allocation of ownership of the **IPRs** to the supplier *(i.e. IPRs generated by the supplier during the PCP)*, by giving an absolute value for the price reduction between the price offered in the tender compared to the exclusive development price *(i.e. the price that would have been quoted were IPR ownership to be transferred to the procurers)]*

in order to ensure compliance with the EU R&D&I state aid framework.

The unit prices quoted for each category of items *(e.g. hourly rates for junior and senior researchers, developers and testers)* remain binding for all phases *(i.e. for the duration of the framework agreement).*

The information provided in the financial section of the tender will be used to evaluate the tenders on the basis of the price award criteria and the on/off award criteria A and C.

More detailed information for the phase 2 and 3 offers will be provided in the call-offs. The price for phase 2 and 3 offers must be based on the binding unit prices in the tender and the price conditions set out in the framework agreement. Where new units/unit prices (e.g. for new tasks or equipment) are subsequently added to the phase 2 or 3 offers, they will become binding for the remaining phases.

Similar price breakdowns will be requested for the call-offs for phase 2 and 3.

The financial compensation for IPRs must reflect the market value of the benefits received (i.e. the opportunity that the IPRs offer for commercial exploitation) and the risks assumed by the supplier (e.g. the cost of maintaining IPRs and bringing the products onto the market).
To ensure that a fair market price is offered, Tenderers must state two prices:

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

- The hypothetical price that they would have quoted if all Intellectual Property Rights, including the ownership of Results under the PCP, would have been fully retained by the Buyers Group and Tenderers would not have the possibility to exploit the Results (the "Virtual Price"); and
- The price that takes into account the fact that they keep ownership of the Intellectual Property Rights attached to the Results under PCP, in accordance with the provisions of the contracts, and that they can exploit these Results (the "Actual Price").

The Actual Price will be evaluated according to the formula:

Weight awarded to Price * (Price lowest tender/Price Tender)

The unit prices quoted for each category of items *(e.g. hourly rates for junior and senior researchers, developers and testers)* remain binding for all phases (i.e. for the duration of the framework agreement).

Since all suppliers will be paid by the Lead Procurer (centralised payments) and as Forum Virium Helsinki is the Lead Procurer in the FABULOS PCP valid Finnish and EU VAT legislation will be applied in the project.

In the Tenders for Phases 2 and 3, the Tenderers must also provide a breakdown of price, as in Phase 1.

The FABULOS financial Section of the Tender has to be submitted by means of Annexes F and G. The information provided in this Section of the Tender will be used to evaluate the Tenders on the basis of the price award criteria and of the compliance criteria.

The Lead Procurer may reject a Tender if it has determined that the submitted price, in combination with other constituent elements of the submission, is abnormally low in relation to the subject matter of the procurement and raises concerns with the Lead Procurer as regards the ability of the Tenderer to perform the contracts. If the Lead Procurer considers that a Tender may be abnormally low, he will request the Tenderer to provide, in writing, details of the constituent elements of the tender, in particular with respect to:

- The economy of the services provided;
- The technical solutions chosen;
- Potential exceptionally favourable conditions available to the Tenderer for the execution of the work;
- The compliance with the provisions relating to employment protection and working conditions in force at the place where the work is performed.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

5. Miscellaneous

5.1 LANGUAGE

All communication (relating to either the tender procedure or the implementation of the contract) must be carried out in English.
Tenders as well as offers for phase 2 and 3 call-offs must be submitted in English.

Deliverables must be submitted in English.

Currently, no special language requirements are foreseen for the field testing in Phase 3. Possible changes in this will be communicated in the call-off documents for Phase 3.

5.2 TENDER CONSTITUTES BINDING OFFER

A signed tender will be considered to constitute a firm, irrevocable, unchangeable and binding offer from the tenderer.

The signature of an authorised representative will be considered as the signature of the tender (and will be binding on the tenderer or, for joint tenders, the group of tenderers).

5.3 UNAUTHORIZED COMMUNICATION — QUESTIONS

The Q&A from the open market consultation can be found on https://fabulos.eu/

For further questions, you may contact the lead procurer Forum Virium Helsinki via email (ask@fabulos.fi) in English until 16.10.2018.

Questions and answers will be presented in an anonymised Q&A document that will be published on https://fabulos.eu/ in English (final version planned for 21.10.2018). For phases 2 and 3, the answers will not be published, but distributed to all suppliers that successfully completed the previous phase.

All other contacts (or attempted contacts) will be considered unauthorised and may lead to the exclusion of your tender.

5.4 CONFIDENTIALITY

Tenderers must keep confidential any information obtained in the context of the tender procedure *(including EU-classified information*[12]*)*.

5.5 CONTRACT IMPLEMENTATION

[12] Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU-classified information.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Successful Suppliers will be requested to sign both a framework agreement and specific contracts for phases 1, 2 and 3.

5.5.1 Monitoring

During each phase, contract implementation will be monitored periodically and reviewed against the expected outcomes *(milestones, deliverables and output or results)* for the phase (see 'Expected Outcomes' in paragraph 2.3).

Each supplier will be assigned a main contact person (their supervisor) from the monitoring team appointed by the procurers. The supervisor of each supplier will be communicated with after the award of the Contract.

There will be regular monitoring meetings between supplier and the supervisor/monitoring team. The supervisor will receive the support of a monitoring team if needed with the necessary expertise. Each meeting or visit will follow the same evaluation criteria and procedures.

For the number of monitoring meetings and the locations see Table of Expected Outcomes in paragraph 2.3. The supervisor, or any party designated by it, is entitled to visit the premises of the supplier and his subcontractor(s).

Each supplier must cover its own costs and thus foresee appropriate personnel and travel budgets in its offer for these monitoring meetings.

The monitoring team will provide regular feedback to suppliers after meetings or visits. Detailed information on the role of the supervisor will be provided after the awarding of the contract.

5.5.2 Payments based on satisfactory completion of milestones and deliverables of the phase

Payments corresponding to each PCP phase will be subject to the satisfactory completion of the deliverables and milestones for that phase.

Satisfactory completion will be assessed by the Technical Evaluation Committee . They will take the final decision on the acceptance or rejection of the milestones/deliverables/tests.

Satisfactory completion will be assessed according to the following requirements:

- if the work corresponding to that milestone / deliverable has been carried out
- if a reasonable minimum quality has been delivered
- if the reports have been submitted on time
- if the monies/resources have been allocated to the planned objectives
- if the monies/resources have been allocated and the work has been carried out

according to the on/off award criteria (place of performance, public funding and R&D definition criteria)
and

- if the work has been carried out in compliance with the provisions of the contract (including in particular verification if the supplier has duly protected and managed IPRs generated in the respective phase).

'Reasonable minimum quality' of a report means that:

- the report can be read by somebody who is familiar with the topic, but not an expert
- the report gives insight in the tasks performed in and the results

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

- the report is made using the end of phase report form or (if applicable) the milestone report form and the requirements of this form have been met

'Reasonable minimum quality' of a demonstration (for phase 2 or 3) means:

- the demonstration can be understood by somebody who is familiar with the topic, but not an expert (for instance, somebody with operational but not technical knowledge)
- the demonstration shows how the innovation works, how it can be used and (if applicable) how it is operated and maintained
- the demonstration is accessible to parties appointed by the procurers, unless these are direct competitors of the supplier

Satisfactory completion in each of the phases does not mean successful completion. A PCP could, for instance, be satisfactorily completed even if it concludes that the innovation is not feasible. See also Section 5.5.4 on "Eligibility for the next phase based on successful completion of the phase".

The assessment will consider the efforts made by Suppliers to take into account the feedback from the supervisor or the monitoring team. The Lead Procurer will approve or reject the submitted deliverables as 'satisfactory' within 30 calendar days of their submission.

Where the Evaluation Committee judges the completion of deliverables or milestones to be unsatisfactory, the Lead Procurer may decide to reduce or withdraw payments for that deliverable and/or may terminate the Contract.

Invoices must be submitted to the Lead Procurer. The details regarding the payments by the Lead Procurer are set out in the Framework Agreement.

suppliers' invoices must provide:

- a **price breakdown** showing the price for R&D services and the price for supplies of products (in order to demonstrate compliance with the definition of R&D in on/off award criterion A, Form D)
- a **price breakdown** showing the location or country in which the different categories of activities were performed *(e.g. x hours of senior researchers in country L at y euro/hour, a hours of junior developers in country M at b euro/hour)* (in order to demonstrate compliance with the requirement relating to the place of performance in on/off award criterion C, Form D).

5.5.3 Payment schedule

Payment for the supplier's Services for each Phase will be made according to the following provisions:

(i) Payment schedule for Phase 1 will be:
100% of the price offered by the Supplier shall be paid by the date in which the Lead Procurer declares the satisfactory completion of Phase 1, as described in Expected outcomes of the Request for Tenders Expected Outcomes.

(ii) Payment schedule for Phase 2 will be:
40% of the price offered by the Supplier shall be paid by the date in which the Lead Procurer

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

declares the satisfactory completion of the Phase 2 Interim Results, as described in Expected outcomes of the Request for Tenders (paragraph 2.3.2):
M2.3) Second Prototype iteration and lab test demonstration
D2.2) Lab test results report

60% of the price offered by the Supplier shall be paid by the date in which the Lead Procurer declares the satisfactory completion of Phase 2.

(iii) Payment schedule for Phase 3 will be:
40% of the price offered by the Supplier shall be paid by the date in which the Lead Procurer declares the satisfactory completion of the Phase 3 Interim Results, as described in Expected outcomes of the Request for Tenders (paragraph 2.3.3):
M3.3) Interim Field test monitoring
D3.4) Field test iteration exercise

60% of the price offered by the Supplier shall be paid by the date in which the Lead Procurer declares the satisfactory completion of Phase 3.

Payments will be made to the bank account provided by the Supplier within 30 days from the date of receipt, by the Lead Procurer, of a correct and approved invoice.

5.5.4 Eligibility for the next phase based on successful completion of the phase

Eligibility for participation in the next phase will be subject to *successful* completion of the current phase.

Successful completion of a phase will be assessed by the assessment committee against the following requirements:

- if all milestones have been successfully completed
- if the R&D results meet the minimum functionality/performance requirements of the challenge description (i.e. the minimum quality/efficiency improvements which the procurers set forward for the innovative solutions to achieve)
- if the results of the R&D are considered to be promising

'Promising' means:

- for phase 1, that the feasibility is convincing
- for phase 2, that the feasibility, the applicability in an operational setting and the potential impact of the solution is convincing

Note that there is a difference between satisfactory completion (requirement for payment) and successful completion (prerequisite for passing from one phase to the next).

Finalisation of phase 3: Possible follow-up PPI procurements

Follow-up PPI procurements for a *limited* set of prototypes and/or test products developed during this PCP procurement *('limited follow-up PPIs')* may be awarded by negotiated procedure *(with invitation to at least 3 potential providers, including those that successfully completed this PCP).*

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Follow-up PPI procurements for a *commercial volume* of the innovative solutions developed in this PCP procurement will be subject to a new call for tenders.

5.6 CANCELLATION OF THE TENDER PROCEDURE

The procurers may, at any moment, cease to proceed with the tender procedure and cancel it. The procurers reserve the right not to award any contracts at the end of the tender procedure.The procurers are not liable for any expense or loss the tenderers may have incurred in preparing their offer.

5.7 PROCEDURES FOR APPEAL

Decisions taken with regard to the selection of Suppliers, awarding them with Phases 1, 2 or 3 or excluding them from the FABULOS PCP Procedure can be challenged by means of an administrative remedy within a period of 5 days upon the formal notification of the decision.

A decision dismissing the appeal could be challenged before the District Court of Helsinki.

Any dispute or claim arising out of or in connection with the execution of the Framework Agreement or of the Phases contracts entered into between the Procuring Partners and the Supplier, shall be heard by the District Court of Helsinki.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Appendix 1 – PCP Principles Horizon 2020

The **PCP is split into three phases** (solution design, prototyping, original development and testing of a limited set of 'first' products or services). Evaluations after each phase progressively identify the solutions that offer the best value for money and meet the customers' needs. This phased approach allows successful suppliers to improve their offers for the next phase based on lessons learnt and feedback from procurers in the previous phase. Using a phased approach with gradually growing contract sizes per phase also makes it easier for smaller companies to participate in the PCP and enables SMEs to grow their business step-by-step with each phase.

The R&D services can cover research and development activities ranging from solution exploration and design, to prototyping, right through to the original development of a limited set of 'first' products or services in the form of a test series. Original development of a first product or service may include limited production or supply in order to incorporate the results of field-testing and demonstrate that the product or service is suitable for production or supply in quantity to acceptable quality standards. R&D does not include quantity production or supply to establish the commercial viability or to recover R&D costs.[13] It also excludes commercial development activities such as incremental adaptations or routine or periodic changes to existing products, services, production lines, processes or other operations in progress, even if such changes may constitute improvements.

The following information provides a generic introduction to the PCP instrument, in accordance with the documentation provided by the European Commission. For a more detailed description of the PCP concept be sure to read the following EU documents: (1) PCP Communication COM/2007/799 and (2) Staff Working document SEC/2007/1668.

Furthermore the PCP is also characterised by the following key **features**:

- **Open, transparent, non-discriminatory approach**

PCP is open to all operators on equal terms, regardless of the size, geographical location or governance structure. There is, however, a place of performance requirement that they must perform a predefined minimum percentage of the contracted R&D services in EU Member States or Horizon 2020 associated countries. In FABULOS' case it is 50%.

- **Sharing of IPR-related risks and benefits under market conditions**

PCP procures R&D services at market price, thus providing suppliers with a transparent, competitive and reliable source of financing for the early stages of their research and development. Giving each supplier the ownership of the IPRs attached to the results it generates during the PCP means that they can widely exploit the newly developed solutions commercially. In return, the tendered price must contain a financial compensation for keeping the IPR ownership. Moreover, the procurers must receive rights to use the R&D results for internal use and licensing rights subject to certain conditions.

[13] See also Article XV(1)(e) WTO GPA 1994 and the Article XIII(1)(f) of the revised WTO GPA 2014.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

Depending on the outcome of the PCP, procurers **may or may not decide to follow-up** the PCP with a 'Public Procurement to deploy the Innovative solutions' (PPI).

However, any subsequent PPI, for the supply of commercial volumes of the solutions, will be carried out under a separate procurement procedure. Suppliers that did not take part in this PCP (or were not chosen to go through as far as the last phase) will thus still be able to compete on an equal basis in any subsequent procurement looking for suppliers to provide a solution on a commercial scale.

More information on PCP can be viewed on the European Commision PCP webpage.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Appendix 2 – Procuring Partners and Preferred Partners

CITY PROFILES AND CURRENT SITUATION

Procuring Partners

This section provides information about the Procuring Partners, which is relevant to potential suppliers (in light of a.o. the field testing). The Buyers Group consists of six cities from smaller towns to capitals, all with different geographical locations and climate. There is also diversity in how the automated driving and field testing are regulated locally, how the process is for obtaining exemptions and how long it takes (from 2 weeks to 6 months), and which stakeholders should be consulted and included in the prototyping and field test phases. Depending on the matureness of each city, there is open data and APIs available for the Suppliers' needs.

For each of the field testing cities there is a description of:

- The city
- The public transport system
- The city's role with regard to public transport
- The city's mobility policy
- Weblinks

For more detailed description of the routes, please see Annex 1 of Tender Document 2: Functional Specifications.

The following information will be found on Annex 2: Local conditions for piloting of Tender Document 2.

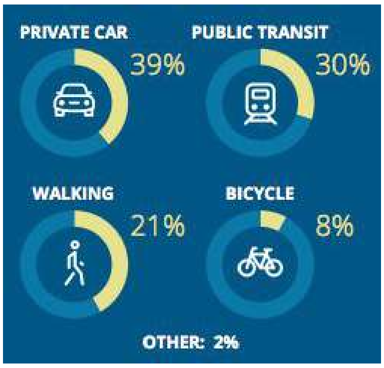
- Relevant data and list of open API's that are related to traffic data in the city
- Information on wireless network capabilities that are expected to be available in Phase 3
- List of the vendor(s) and technology(/ies) currently used in (smart) traffic lights
- Information on the availability of route-wide remote camera surveillance
- Description of the expected size and type of the market after the end of the PCP process.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

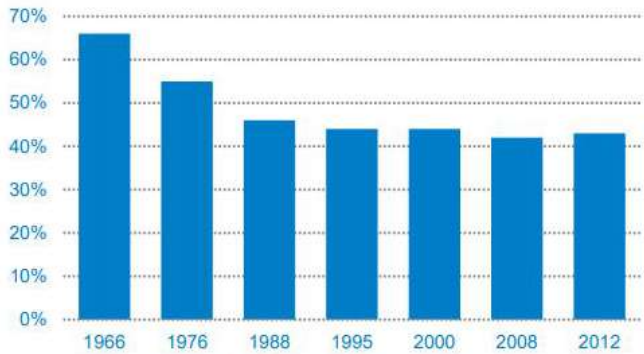
FINLAND: CITY OF HELSINKI

Inhabitants: Helsinki 642 045 (2017), Helsinki Capital Region 1,456,619 (2016)

Size: Helsinki 215,12 km², Helsinki Capital Region 791 km2
Modal split:



Journey Modal Split, Deloitte 2018



Share of public transport, HSL 2018

The public transport system:
HSL procures bus, tram, metro, commuter train and ferry services for the Helsinki Capital Region. In 2016 there were 292 bus lines that HSL procured from Helb, Nobina, Veiolia, Pohjolan LIikenne and other smaller providers (around 10 in total). Metro (2 lines) and trams (13 lines) are provided by HKL (Helsinki City Transport), commuter trains (14) by VR Group (State owned railroad company) and ferry services (2) by HKL/SL. Regarding the fleet, there were 1,424 buses, 46 metro trains (2018), 94 trams, 70 commuter trains and 4 ferries.

Role with regard to public transport:
Forum Virium Helsinki (the Lead procurer of FABULOS) is an innovation unit within the Helsinki City Group. It develops new digital services and urban innovations in cooperation with companies, the City of Helsinki, other public sector organizations, and Helsinki residents.

Helsinki Region Transport (HSL) is a joint local authority consisting of Helsinki and 8 other municipalities, working since 2010. HSL plans and organizes public transport in the region and improves its operating conditions, procures bus, tram, Metro, ferry and commuter train services, is responsible for the preparation of the Helsinki Region Transport System Plan (HLJ) and approves the public transport fare and ticketing system as well as ticket prices. HSL is also responsible for public transport marketing and passenger information, and organizes ticket sales and is responsible for ticket inspections

Licenses and permits for vehicles are given by Finnish Road Safety Association.

Mobility policy:

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Helsinki City Strategy is to be “The most functional city in the world”. Helsinki also aims to become a carbon neutral city by 2035. Achieving this goal will require the City to conduct a variety of trials in order to find the best emission-free solutions. In the process of doing so, the City also aims create the most effective transport services in the world. For residents, this will mean carbon neutral transport options and easy-to-use services, while companies can look forward to a world-class test environment for smart traffic. These development efforts will also provide the City of Helsinki with a competitive advantage and the science community with a fruitful research environment. There have already been several robot bus trials conducted in Helsinki. Forum Virium Helsinki has been a partner and/or involved in many of them prior to FABULOS.

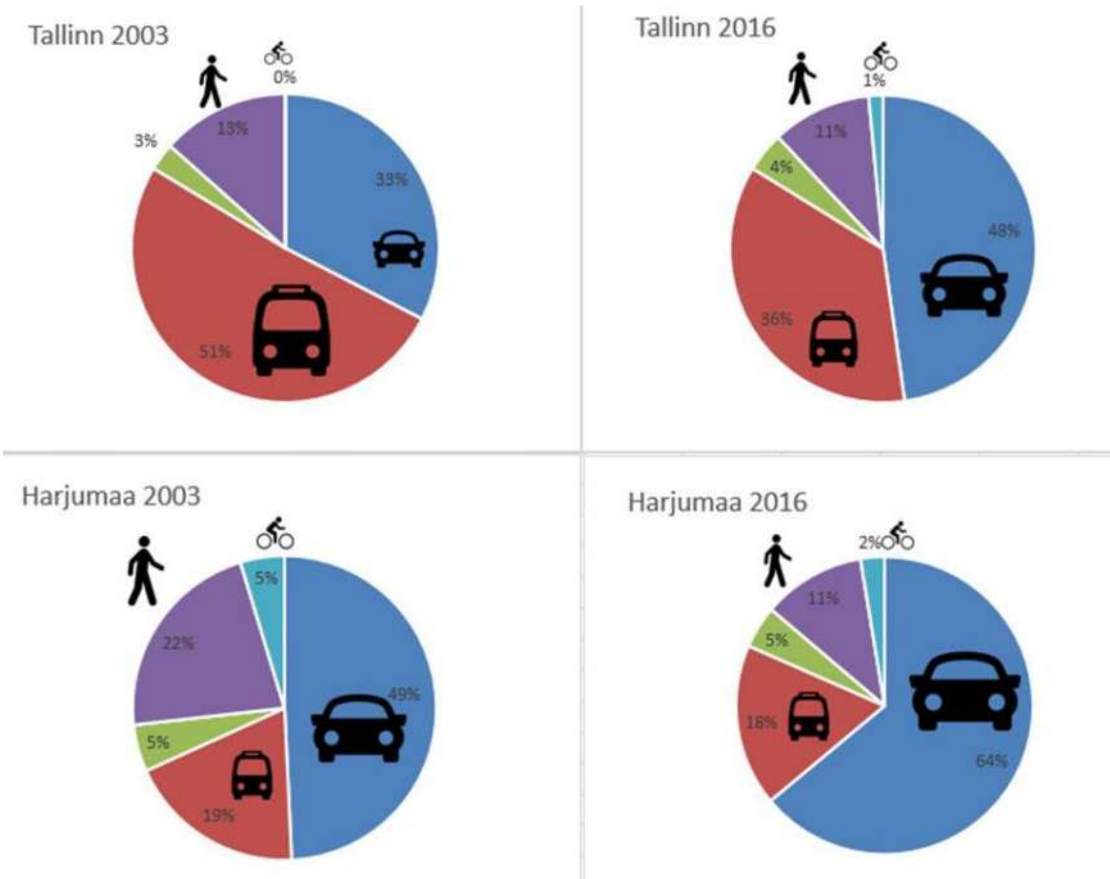
Relevant weblinks:
Forum Virium Helsinki: https://forumvirium.fi/en/
Helsinki City: https://www.hel.fi/helsinki/en
HSL: https://www.hsl.fi/en
HKR: https://www.hel.fi/hkl/en

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

ESTONIA: CITY OF TALLINN

In Estonia, the FABULOS procurement partner (member of the Buyers Group) is the Ministry of Economic Affairs and Communications. The field testing in Phase 3 will take place in the capital of Estonia, Tallinn.

Tallinn has about 450,000 inhabitants, size 159,3 km^2. Modal split in Tallinn and in Tallinn greater area (Harjumaa):



The public transport system

Public transit is free. There are buses, trolley buses and trams in the city and also trains servicing the city and its greater area operated by the state owned firm Elron. Here are the main public transit lines and their schedules, operated by Tallinn mainly: https://transport.tallinn.ee/#tallinna-linn/en.

Role with regard to public transport

Tallinns public transport is procured by the city of Tallinn and is operated by Tallinna Linnatranspordi AS, a city owned company: http://www.tallinnlt.ee/en/. Licenses and permits are

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

given by Estonian Road Administration, which is a subordinate institution to the Estonian Ministry of Economic Affairs and Communications (the procuring partner, henceforth MKM). MKM is responsible for all the legislation regarding transport (e.g.) in Estonia and also responsible for the management and allocation of funds in transport sector (both from state budget and e.g. EU financing).

Mobility policy

Estonia has a quick-charging network that encompasses all of Estonia geographically (40-60 km distance between charging points). See more: http://elmo.ee/charging-network/

Relevant weblinks:

Organisation, MKM: https://www.mkm.ee/en

Pilot city: https://tallinn.ee/eng/

Road Administration (authority): https://www.mnt.ee/eng.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

NORWAY: GJESDAL MUNICIPALITY

Gjesdal is a mountainous district, on the west coast of Norway, approximately 30 km southeast of Stavanger. The population of Gjesdal has passed 12 000, and is spread over an area of 640 km². Ålgård is the county's town centre, with 10000 inhabitants. Gjesdal has one of Norway's youngest populations. Gjesdal strives to put its inhabitants first, whilst also developing commercial activity. We aim to provide a holistic service for the user, where internal departments collaborate to provide best service and with the most effective use of resources. Our vision is "We go further" and our main objective is to ensure that Gjesdal is a great place to live, work and do business. The main road E39, passes through Ålgård. E39 leads to Sandnes and Stavanger in the north and to Kristiansand in south. The road is used by thousands of motorists and lorries every day, as it is the main connection between the east coast and west coast of Norway. The only public transport system from/to Gjesdal is busslines operated by Kolumbus. On this link you can se on a real time map were the bus are right now (busline 23 goes from Sandnes to Ålgård)

Role with regard to public transport:

Kolumbus is a public transport provider own by Rogaland County Council and is responsible for bus and boat traffic in Rogaland County wich Gjesdal is a part of. The transport by bus and boat itself is operated by companies on contract with Kolumbus.

Gjesdal has Smart City as development strategy and mobility is one of the main targets. After building a new city center the increase of commercial and shopping activities has resulted in significant mobility challenges and increasing parking needs. We want to reduce local traffic and parking needs by offering the inhabitants alternative mobility solutions. This can be electrical bicycles, car pooling, minibuses for special routes, a modern bus stop with free wi-fi and the possibility for parking bike safely in the city center. We also have charging points for electric cars in the city senter. Shared, electric and autonomous mobility will fit us really well.

Open data for Kolumbus on this link. Gjesdal has some open data, but not on public transport at the moment. As the open data is available, it will be published on this web page.

Relevant weblinks:

Gjesdal municipality www.gjesdal.kommune.no
Smart Gjesdal www.gjesdal.kommune.no/tjenester/smart-city/
Kolumbus https://www.kolumbus.no/en/

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

GREECE: LAMIA MUNICIPALITY

Lamia Municipality is the capital of the Region of Sterea Ellada, located in the center of Greece and a key-reference point on an administrative, commercial and industrial level. The Municipality is home to 74.720 citizens and covers a geographical area of more than 1 million m^2 area. Moreover, Lamia as a Municipality supports a wide administrative and business zone of influence which covers neighboring Municipalities, resulting in having a population equivalent of 200.000.

The public transport system:

A sophisticated fleet of city buses, a rich network of pedestrian and bicycle routes complemented by public and private parking infrastructures, support the daily transportation needs of citizens. In general, the city center is characterized by a challenging and aged road infrastructure with narrow passages, steep slopes, lack of adequate side roads and a wide accumulation of services, resulting in heavy traffic. However, traffic is substantially normalized outside the city center.

Citizens' transportation needs are addressed by 20 public bus lines and 2 mini-bus lines jointly operated by the Municipality and a local private transport operator. The utilized bus fleet is comprised by 26 vehicles with 40 seats and 4 mini busses of 30 seats, offering low-cost transportation services to citizens for tickets priced from 0,6€ to 1,70€ and connecting the city center with different remote city locations and nearby villages.

Moreover, Lamia supports a 7.5 km network of bicycle paths, which connects the city's center with its southernmost districts. The network includes two different paths for bicycles and pedestrians, flowerbeds, sidewalks and lighting poles. Public parking spaces and several public transport hubs are strategically located all along the network, thus enhancing the transportation experience of citizens. Finally, the Municipality operates an extended controlled parking system including private and public parking infrastructures to satisfy daily citizens and visitors parking needs.

On average and according to the latest measurements, 8.000 tickets per day and almost 3 billion tickets per year are issued for the fleet of public busses navigating the city. Additionally, traffic measurements in key city center crossroads during workdays and peak hours showcase a traffic of 700 private and public vehicles per hour on average.

Role with regard to public transport:

The Municipality has full administrative and financial jurisdiction, responsibility and independence in managing, designing, procuring and purchasing public transport infrastructure under the provisions of Greek procurement laws and directives for Local/Municipal authorities. In this sense, the Municipality and its different departments – mainly the Departments of Procurement and Financial Services – retain a central role in the underlying procurement procedures and are engaged in all different procurement phases according to distinct procurement requirements.

The Municipality is committed to the steady modernization and digitalization of its public transportation services and the systematic minimization of private vehicles usage within the city center. In this context, the Municipality has been investing throughout the last years on innovative solutions, knowledge economy and smart infrastructures. The long-term vision is to formulate a best practice example in Greece for smart city transport solutions, dealing with

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

issues of traditional transportation issues, enhancing the quality of provided services and finally supporting the Municipal policy of minimizing the usage of private cars while augmenting the exploitation of shared, electric and autonomous mobility. This vision is backed by several regional and local programming financial instruments, such as the Regional Operational Programme and the Municipal Sustainable Urban Development Plan, both European Regional Development Fund based.

In terms of transport data, the Municipality retains a network of traffic and environmental monitoring sensors, collecting critical data throughout the city and its surroundings. Accordingly, relative open data repositories are available for mobility and navigation data, points of interest, events, routes and transportation accessibility information among others.

Relevant weblinks:
Lamia Municipality website: http://lamia.gr/,
Local private transport operator website: http://www.astikoktellamias.gr/

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

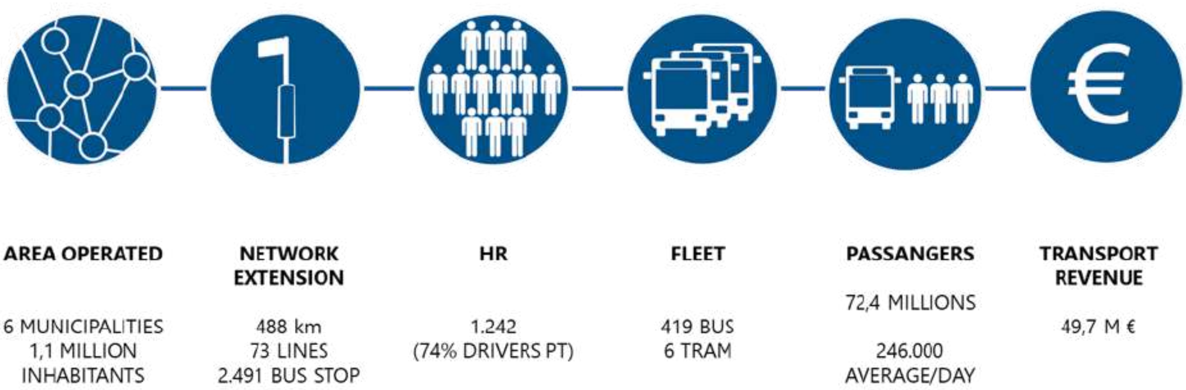
PORTUGAL: STCP, CITY OF PORTO

In the city of Porto, the FABULOS procurement partner (member of the Buyers Group) is the public transport operator of the city, STCP - SOCIEDADE DE TRANSPORTES COLECTIVOS DO PORTO SA. The field test in Phase 3 will take place in the zone of the city called ASPRELA where the main university campus of the city is located. There is located, also, one of the largest hospitals in the region and one of the main transport interfaces of the city.

Porto is the main city in the north of Portugal and has about 238 000 inhabitants, in an urban area of 41.4 km2. The metropolitan area of Porto, the region where STCP operates, is composed of 6 municipalities with more then 1,1 million inhabitants and has 562,3 km2.
Embracing the Douro River, Porto is one of the oldest cities in Europe, with a historical center classified by Unesco since 1996 as a World Heritage Site. The city, whose name is universally known thanks to the famous Port Wine, preserves a remarkable historical patrimony, well visible in its medieval, baroque, neoclassical and romantic monuments. With its ever-present historical and cultural heritage, Porto has modernized itself through the research centers of its University, designed bold public spaces such as the Casa da Música, the Serralves Contemporary Art Museum, the Faculty of Architecture, putting the city on the routes of Modern Architecture. At the same time, it has created new infrastructures and accessibilities capable of hosting events worldwide. Porto Airport is today an important development factor for the city, as well as cruises and even tourism that arrives by road. The excellent transport network, the remarkable hospital network that it possesses, the security levels it presents and its social and cultural dynamics make Porto today a welcoming and attractive city, having been voted Best European Destination in 2014 and 2017.

The public transport system:

Transport in the Oporto Metropolitan Area is characterized by the use of the various modes of transport: road passenger transport network, suburban railway and light rail lines.
The STCP network occupies a leading position in collective road transport in the Metropolitan Area, with the public transport company operating a network with 73 lines (59 line buses, 11 in the network of dawn and 3 tram) with a network extension of 480 km, with 2 465 stops and a fleet of 419 buses and 6 historical tram, being used by 72.4 million passengers (2017).



This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Role with regard to public transport:
STCP have the mission to provide a public transport service of passengers in the metropolitan area of Porto (AMP), in conjunction with concerted road operators, rail and light rail, contributing to the effective mobility of people, providing a competitive alternative to individual transport and generating, by his private activity, social and environmental benefits within a framework of economic rationality and pursuit of continual improvement of its performance.

Relevant weblinks:
STCP: http://www.stcp.pt/en/travel/
AMP: http://portal.amp.pt
CM Porto: http://www.cm-porto.pt/
AMP-Autoridade Transportes http://autoridade.amp.pt/pt-pt/

NETHERLANDS: CITY OF HELMOND

The City of Helmond is a medium-sized city in the southeast of the Netherlands. It is part of the Brainport region which is one of the three most important economic areas in the country. Innovation is the key-word here and for Helmond the focus lies on innovation in the fields of Automotive and (Smart) Mobility.

- Population: 90,581
- Density : Approx. 1700/km²
- Area Land: 53.23 km²
- Area Water 1.52 km²

The Brainport region consists of leading high-tech industries, amongst them many companies that are active in the automotive/mobility sector and many innovative SMEs. Next to industry, the region is known for its focus on research and development at the Eindhoven University of Technology (TU/e), TNO, the High Tech Campus and the Automotive Campus. The automotive industry is a spearhead in the regional economic development policy and program.

Helmond is home to the Automotive Campus, where over 35 companies work on innovative mobility solutions. With the Automotive Campus, Helmond presents itself as knowledge and innovation centre for mobility solutions and as "City of Smart Mobility". Though a relatively small city Helmond receives internationally recognition for its ambitious goals and the realisation of innovative mobility solutions in the city. The Brainport region will host the ITS Europe 2019 Congress, where the active role of Helmond as a Living Lab for ITS will be showcased.

The public transport system
(Inter)national and regional:

The city of Helmond is well connected to the rail network by four train stations. Every hour, two sprinters stop at each station. The intercity train also stops twice at Helmond Central.

- Station Helmond central

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

- Station Brandevoort
- Station Helmond 't hout
- Station Helmond Brouwhuis.

In addition, we have a number of regional bus lines for direct connections with surrounding areas.

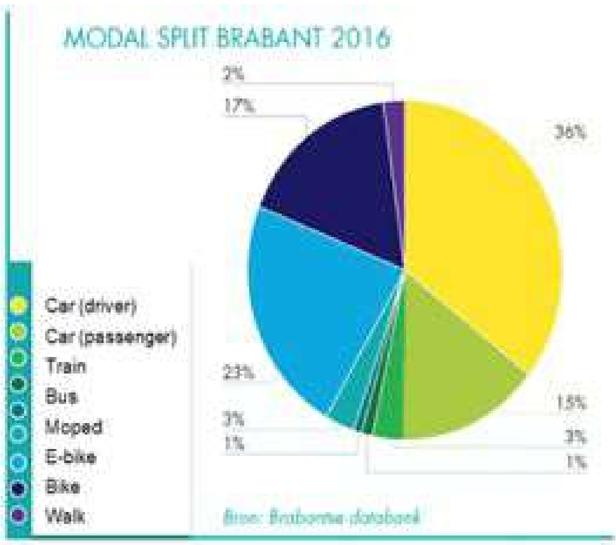
Local public transport:

Many medium-sized cities face the same problem when it comes to low occupancy rates in buses during off-peak hours to justify a dense and frequent public transport system. Some specific reasons for this are in Helmond: a very good network of cycle paths, relatively small distances from residential areas to the nearest train station, and not being a city with many students.

To provide a good public transport system that fits well with the needs of the city, the public transport authority, public transport operator and Helmond are testing a new system called Bravo*flex*. Based on the concept "Mobility as a Service (MaaS)", we have introduced a system with minibuses without fixed lines or a timetable. Bus travellers can now order a bus trip via an app on the smartphone or by calling. The app gives real time information of the expected time of arrival. The minibus picks up the traveller (at the departure bus stop) and drives him directly to the bus stop nearby his destination. Part of this pilot is also research into possible integration between public transport and transport target groups.

With this new on demand public transport service, the new approach based on MaaS and the integration with target groups, we give the public transport in Helmond a positive boost.

By improving our local public transport, we also want better accessibility for our hotspots, such as the Automotive Campus. And we are convinced that this is possible with solutions as requested in the FABULOS pre commercial procurement, where a fleet of self-driving minibuses can take care of the last mile.



And with a proof of concept, it seems possible to expand FABULOS buses and replace the current fleet over time.

<u>Modal split and local public transport:</u>

Bus transport is a relative small part of the total mobility in the province of North Brabant (i.e. 1.4 % out of all movements and 1.8 % of all kilometers). This is the average for the whole of Brabant; in the city of Helmond this percentage will even be lower.

Role with regard to public transport:

The roles are divided as follows:

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

- Province North Brabant is public transport authority
- Hermes is public transport operator
- Municipality Helmond is road authority and has great interests in good public transport

The province of North Brabant determines (via public tendering) which company will provide the public transport in this region. Public transport companies can participate in the tender and the one with the best bid will obtain the concession. The current concession commenced in December 2016 and has a term of 10 years. There is space created in the contract for further development of innovative concepts. Fabulos and Bravo*flex* are great examples of this. The road authority is responsible for the infrastructure and is of course an important partner for the province during the tender.

Zero-emission:

The province, our region and the city of Helmond are committed to reducing polluting emissions. At the start of the concession Hermes introduced 43 electric buses in Eindhoven. In 2024 the complete bus fleet must drive with zero emission. This will also apply to the minibuses in Helmond.

Relevant weblinks:

- The city of Helmond: www.helmond.nl
- Brainport Eindhoven: www.brainporteindhoven.com
- Automotive Campus: www.automotivecampus.com
- Province North Brabant: www.brabant.nl
- Public Transport operator Hermes: www.hermes.nl
- Public Tranport information North Brabant: https://www.bravo.info/
- Public Transport Travel information: www.9292.nl/en

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

PREFERRED PARTNERS

To ensure the further validation of the pre-commercial procurement challenge definition, and more direct future market up-take for the solution, the FABULOS project has set up a "Preferred partner" group consisting of cities and/or public procurers. The following entities are participating as preferred partners with an interest in the PCP, but without being part of the buyers group or giving in-kind contributions for carrying out the PCP:

City of Copenhagen (DK)
MOVIA - regional transportation company for Greater Copenhagen region (DK)
City of Torino (IT)
City of Ghent (BE)
City of Trikala (GR)
City of Delphi (GR)
City of Fundão (PT)
City of Kongsberg (NO)
City of Velje (DK)
City of Vantaa (FI)
Region of Stavanger (NO)
Porto Digital (PT)
City of Tuusula (FI)
City of Buenos Aires (AR)

The Preferred Partners provide information and opinions to help the procuring partners' decision-making, for example, by participating in some of the activities of the Open Market Consultation. They promote the results of the FABULOS pre-commercial procurement process to the policy community and other adopter cities. They also promote the FABULOS request for tenders to potential suppliers in their constituencies. They will keep the FABULOS partners informed of any implementation of self-driving buses in their municipalities and the results..

Preferred Partners are entities that are neither members of the buyers group, nor third parties providing in-kind contributions to the FABULOS project, but that have a special interest in closely following the FABULOS pre-commercial procurement. They are interested in implementing the solution coming out of FABULOS in their own cities. They will not have rights to results or IPRs. The Preferred partners will receive some funding to cover travel costs up to two FABULOS events, of which one is the Final conference. During the Final Conference demonstrations of the final solutions will be given. A special workshop session for Preferred Partners will be hosted, with a view to a potential common PPI.

Also, a FABULOS Preferred Partners event will be organised in the margins of the Transport Research Arena Conference in April 2020 in Helsinki. This will also be the time that the pilot in Helsinki is ongoing, so Preferred Partners will get a demonstration of the solution and a peek behind the scenes.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Appendix 3 – Electronic submission of the FABULOS Tender

Tenders have to be submitted electronically via the fabulos.eu website. On the Request for Tenders page, all documents will become available on TED and for download on the fabulos.eu website 1.9.2018.

There will be a separate page where Suppliers can submit their offers by uploading them. No pre-registration or logging in is needed for this. No special software is needed.

- The total size for all files together cannot exceed 100MB
- Give files a clear name (e.g. FABULOS_FormA_...);
- Make sure to follow the instructions on the designated page;
- Make sure your upload is complete.

When in doubt, contact the helpdesk via ask@fabulos.eu, but be sure to do this reasonably in advance.

Tenderers are asked to have an authorized representative sign each of the Submission Forms (Forms A through G) separately, either manually ("blue ink") in the foreseen signature box (when provided) or electronically, and upload them separately.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Appendix 4 - Scoring Model for the Award Criteria

Overall scoring model
This Appendix contains the scoring model that will be used by the evaluators to assess and score the extent to which a Tender is meeting the award criteria.

Score	Textual description	
0	**Nonexistent**	None of the aspects of the requirement are met.
1	**Very weak**	Multiple important aspects of the requirement are missing.
2	**Weak**	Multiple aspects of the requirement are present, but the provided explanation may not convince.
3	**Good**	All important aspects are present.
4	**Very good**	All important aspects are present and the provided explanation is very convincing.
5	**Excellent**	There is significant added value to the required feature, which is described very convincingly.

Table 13. Scoring model

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371.

Appendix 5 - Scoring Model for the Price

1. Scoring of the Total PCP Price
The Price will be evaluated using the formula below:

Points awarded = Weight awarded to Price * (Price lowest tender/Price Tender PCP)

Price Tender PCP = is the Actual Price that the Tenderer has submitted for the Total PCP (Phases 1, 2 and 3).

Weight awarded to Price = the maximum points the Tenderer can get on the Price award criterion (see Overview of the award criteria of the Request for Tenders)

2. Scoring of the Price for PCP Phase 1
The Price will be evaluated using the formula below:

Points awarded = Weight awarded to Price * (Price lowest tender/Price Tender Phase 1)

Price Tender Phase 1 = is the Actual Price that the Tenderer has submitted for Phase 1.

Weight awarded to Price = the maximum points the Tenderer can get on the Price award criterion (see Overview of the award criteria of the Request for Tenders)

Note that the Actual Price will be taken into account (see Form F), and not the Virtual Price, for the evaluation.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Appendix 6 - Scoring Model for Outcome of the Phases

This Appendix contains the scoring model that will be used by the evaluators to assess the successful completion of Phase.

Score	Textual description	
0	**Nonexistent**	None of the aspects of the requirement are met.
1	**Very weak**	Multiple important aspects of the requirement are missing.
2	**Weak**	Multiple aspects of the requirement are present, but the provided explanation may not convince.
3	**Good**	All important aspects are present.
4	**Very good**	All important aspects are present and the provided explanation is very convincing.
5	**Excellent**	There is significant added value to the required feature, which is described very convincingly.

Table 14. Scoring guide

Quality of the end of Phase report of Phase 1, 2 or 3
For the End of Phase report, each of the following areas is assessed:

1. There is evidence that the work has been carried out completely and diligently;
2. The results are good and consistent with the original Tender;
3. There is a clear potential for further development;
4. The report is well written with the appropriate level of detail.

On each of these areas a tender can score a maximum of 2,5 point (adding up to a total of 10 points).

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

Appendix 7 - Table of page limits

Follow the page limits specified below for the submission forms of your Tender, unless otherwise specified.

Submission Forms	Page limits of Submission Forms
Form A - General Tender Submission Form	15
Form B - Exclusion Criteria (declaration)	5
Form C - Selection Criteria	20
Form D - Compliance Criteria (declaration)	5
Form E - Technical Offer	40
Form F - Financial Offer & Cost Breakdown	25
Form G - Financial Offer Phase 1	3

Table 15. Page limits

Tenderers will use a minimum font size of 10. Use a minimal line spacing of 1.

Tenders exceeding a page limit: words and/or pages in excess of the specified limit may not be considered further. Tenders not complying with the minimal font and spacing size may be eliminated.

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Appendix 8 - End of Phase Reporting [sample]

Below is a sample template of an End-of Phase Report to be used throughout the project to document progress. It is provided here as an example for your information. The actual form will be provided with the requirement specification of the relevant phase.

Buyers group *Complete this box only one time with the joint conclusions from all procurers in the buyers group*
1. Procurement need *Describe briefly (in a way that is suitable for publication purposes):* **The problem / challenge** you were trying to address with the procurement **What type of innovative solutions and which functionality / performance / price requirements** you requested in the tender specifications (specify the minimum and target quality / efficiency improvements that you wanted the innovative solutions to achieve).
2. Impact on public sector modernization *Describe briefly (in a way that is suitable for publication purposes):* **To what extent the innovative solutions managed to meet the procurement need so far** (which tender requirements were the innovative solutions not able / able / more than able to meet)? For PCPs, specify whether all participating contractors managed to complete the previous phase successfully Did their solutions all meet the procurement need / the tender requirements? What is the current impact of the innovative solution on end-users? **What level of quality / efficiency improvements do the innovative solutions enable to achieve** (use measurable indicators to quantify the impact achieved on the operation of your public service, e.g. *25% reduction in maintenance costs, 30% reduction in mortality rate of patients in your hospital*)

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

3. Other benefits obtained

Describe briefly any other benefits obtained from the procurement, not only for the public procurers involved but also wider benefits for society (in a way that is suitable for publication purposes), e.g.:

Reducing vendor lock-in: *e.g. the procurement delivers more open (standardised) solutions and/or opens a route to the market for new innovative players which creates a more competitive supply chain*

Wider benefits to society: *e.g. contribution to CO2 reduction, improved public safety / health*

Contribution to growth and jobs: For PCPs, specify the percentage of the R&D that the contractors actually performed in the Member States or countries associated with Horizon 2020. For PPIs, specify the percentage of the total PPI contract value that was awarded to contractors from Member States or countries associated with Horizon 2020.

Triggering other innovation procurements: This PCP/PPI triggered management commitment to start new innovation procurements in the future in organisations xyz

Other benefits / lessons learnt: *complete if applicable*

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

4. Scalability — Wider deployment

Describe briefly (in a way that is suitable for publication purposes):

How easy it would be for other procurers to deploy the solutions resulting from the procurement (which parts of the solution are generic /can be replicated by other procurers across Europe versus which parts would still need adaptation / modification to other markets etc.)

What actions did you already take to help diffuse the innovative solutions to wider markets e.g.

- did you / the suppliers in your procurement contribute to standardisation
- did you / the suppliers publish results / lessons learnt of the procurement
- did you require the solutions for your procurement to be based on open interfaces / open source?
- did your dissemination activities promote results / impacts achieved to other procurers?
- did you help the suppliers to go for wider commercialisation of the innovative solutions *(e.g. via joint supplier-procurer presentations of the solutions/impacts at trade fairs, actively acting as first customer reference to other customers, introducing the suppliers to investors, etc.)*
- at the end of the project: did you update the initial tender specifications with the lessons learnt during the procurement and did you publish these updated tender specifications so that other procurers can use them in future procurements?

Which aspects of the initial tender specifications *(in particular functionality / performance / price requirements)* you would change / update after this procurement based on the lessons learnt, to make sure that later procurements that go for wider deployment would run as smoothly as possible

Contactors

For PCPs: complete this box for each contractor that was awarded a PCP Phase 1, 2 or 3 contract. For PPIs: complete this box for each contractor that was awarded a PPI contract

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

1. The innovative solution

Provide a short description (that is suitable for publication purposes) of:

The innovative solution (in its current form)

Where exactly lies the innovativeness in the solution: In which ways and to which extent does the solution go beyond what existing solutions can achieve

The degree of innovation: indicate if your innovative solution is (a) a totally new product / service / process / method; (b) an improvement to an existing product / service / process / method; (c) a new combination of existing products / services / processes / methods (d) a new use for existing products / services / processes / methods)

2. Commercialisation success

Provide a short description (mark parts that are not suitable for publication purposes) of:

How mature is the innovative solution in terms of its readiness to commercialise widely: Which steps towards wide scale commercialisation have been completed by now *(don't forget: IPR protection, certification, CE marking, attracting additional investors to grow the business, setting up sales / distribution channels / marketing activities to expand sales to other countries etc.).*

What is the current commercialisation success of the solution: *e.g. awards / other forms of recognitions obtained, sales / increase in market share already achieved, licensing agreements already concluded, collaboration agreements with other partners (e.g. retailers) to commercialise the solutions already signed, additional investments attracted to further commercialise the solution*

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 780371

3. Other benefits obtained

Provide a short description (mark parts that are not suitable for publication purposes) of any other benefits that you obtained from participating in the procurement, e.g.

Getting easier access to (a new segment of) the public procurement market (did the procurement enable you to work with procurers/end-users that you were not working with beforehand)

Growing your business across borders and/or to other markets *(e.g. private markets)* thanks to the first customer references provided by the procurement

Shortening the time-to-market for your innovation thanks to early customer/end-user feedback

Other benefits / lessons learnt: complete if applicable

4. Business growth

Provide a short description (mark parts that are not suitable for publication purposes) of:

How much has your business already grown during the procurement

In terms of (a) personnel growth; (b) turnover growth; (c) growth in market share etc.

What are the prospects to grow your business via wider commercialisation of the solution:

- how large is the potential market for your solution? is it a growing / steady / declining market?
- by when can commercialisation start (now / in 1 / in 3 / in 5 / in more than 5 years)
- is competition patchy (no major players) / established (but no comparable offering) / fierce

Which future steps do you plan to take to further grow your business *(e.g. attracting additional investors to grow your business, mergers / acquisitions / joint ventures / spin-offs / IPO, setting up sales / distribution channels / marketing activities, expanding to other countries etc.)*

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371

5. Final remarks *(not for publication purposes, to assess how further EU support could best help you)* **What are remaining bottlenecks to commercialise your solution** (e.g. certification, legislation etc.) **What type(s) of assistance do you need to address those bottlenecks and grow your business** / commercialise your solution more widely (e.g. EU regulation on x, finding investors, IPR help etc.) **How important was the procurement for your business** (w/could you have done it on your own?)

Table 16. Sample End of Phase Report template

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Appendix 9 - Project abstract for Phase 1 [sample]

Contactor Details	Type/ size of legal entity	Place of performance of contract activities	Logo
Main contractor Name legal entity Address legal entity Name contact person Phone nr contact person E-mail address contact person	SME, larger company, natural person, university / research institute, other	% of contract value allocated to main contractor: [complete] % % of activities for the contract performed by the main contractor in EU Member States or countries associated with Horizon 2020: [complete] %	Main contractor logo
Other consortium member(s) (if applicable) Name legal entity Address legal entity Name contact person Phone nr contact person E-mail address contact person	SME, larger company, natural person, university / research institute, other	% of contract value allocated to contractor [x]: [complete] % % of activities for the contract performed by contractor [x] in EU Member States or countries associated with Horizon 2020: [complete] %	Other contractor(s) logo(s)

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371.

Subcontractors (if applicable) Name legal entity Address legal entity Name contact person Phone nr contact person E-mail address contact person *Complete as many times as there are subcontractors*	SME, larger company, natural person, university / research institute, other	% of contract value allocated to subcontractor [x]: [complete] % % of activities for the contract performed by subcontractor [x] in EU Member States or countries associated with Horizon 2020: [complete] %	Subcontractor (s) logo(s)
Project abstract ***(+/- 1000 characters maximum)*** [Add an abstract of the winning tender, giving a brief project description agreed with the contractor that is suitable for publication purposes]			
Previous EU funding Is the project based on / a continuation of R&D activities that were previously funded by the EU?: YES/NO If yes, identify this EU funding: [name EU funding programme] — [project name] — [grant number]			

Table 17. Sample Project Abstract template

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 780371